Manufacturing and Service Operations MSO 40.012

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Wrap-up week 1

Manufacturing and Service Operations

MSO transforms inputs into outputs to create value, acting as the core engine within organizations.

Operations Management Importance

OM enhances productivity, optimizes costs, and strengthens competitive advantage in organizations.

Evolution of Operations

Operations evolved from vertically integrated systems to modern Industry 4.0 digital environments.

Three Operational Horizons

Operations cover tactical, executional, and strategic responsibilities essential for business success.

Tactical Operations

Tactical tasks include demand forecasting, production planning, quality assurance, and inventory management.

Executional Activities

Executional operations focus on scheduling, procurement, buying decisions, and cost control daily.

Strategic Decisions & Trade-Offs

Strategic choices involve facility location, capacity planning, risk management, and balancing operational trade-offs.

Wrap-up week 1

Product Characteristics Impact

Product features like shelf life, volume, ... influence manufacturing and supply chain strategies significantly.

Segmentation and ABC Analysis

ABC analysis classifies SKUs by revenue and importance, guiding resource allocation effectively.

Production System Frameworks

MTS, MTO, ATO, and ETO frameworks cater to different customer needs and operational constraints.

Late Differentiation Strategy

Late differentiation reduces inventory and improves responsiveness by postponing customization.

Production Line Types

Organizations select production line types based on product variety and volume to optimize efficiency and customization.

Customization and Volume

Project-based systems handle highly customized low-volume work, while production lines focus on high-volume low-variability manufacturing.

Impact of Mix and Volume

Mix and volume dynamics influence machine layout, resource specialization, process flow, and cost structure.

Definition of Capacity

Capacity is the maximum output a system can achieve considering constraints like labor, equipment, space, and time.

Bottleneck Theory

Improving bottleneck resources increases overall system output, while enhancing non-bottlenecks does not affect capacity.

Balance between load and capacity: cost impact

The actual production capacity is 70 000 units/year. Adding a new production line will increase capacity up to 100 000 units per year but will also increase the fixed cost.

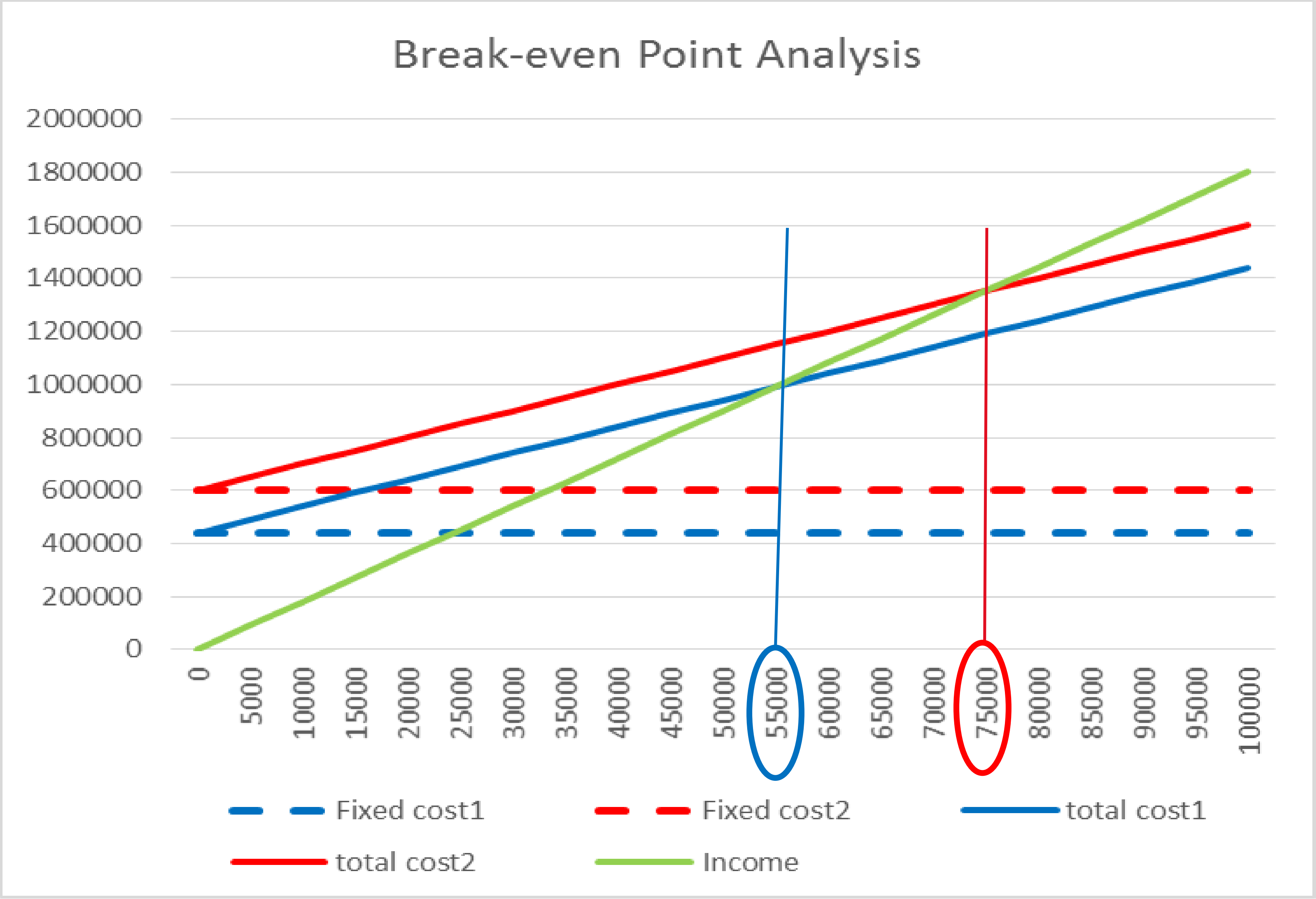
	Before the new production line	After the new production line
Fixed Costs	▪ 440 K€	▪ 600 K€
Direct cost per unit	▪ 10 €	▪ 10 €
Unit Selling Price	▪ 18 €	▪ 18 €
Effective capacity	▪ 70 000	▪ 100 000

What are the impacts on the break-even point and on the Net Income of the year

Balance between load and capacity: cost impact

	Before the new production line	After the new production line
Fixed Costs	▪ 440 K€	▪ 600 K€
Direct cost per unit	▪ 10 €	▪ 10 €
Unit Selling Price	▪ 18 €	▪ 18 €
Total capacity	▪ 70 000	▪ 100 000
Total of direct cost	▪ 10€ x 70 000 = 700 K€	▪ 10€ x 100 000 = 1000 K€
Total Income	▪ 18€ x 70 000 = 1 260 K€	▪ 18€x 100 000 = 1 800 K€
Margin	▪ 1 260 – (700 + 440) = 120 K€	▪ 1 800 – (1000+ 600) = 200 K€
Break-even point	▪ 440 / (18-10) = 55 000	▪ 600 / (18 -10) = 75 000

Potential increase of the margin: 80 K€ (66%)
Increase of the break-even point: 36 %.



After new line

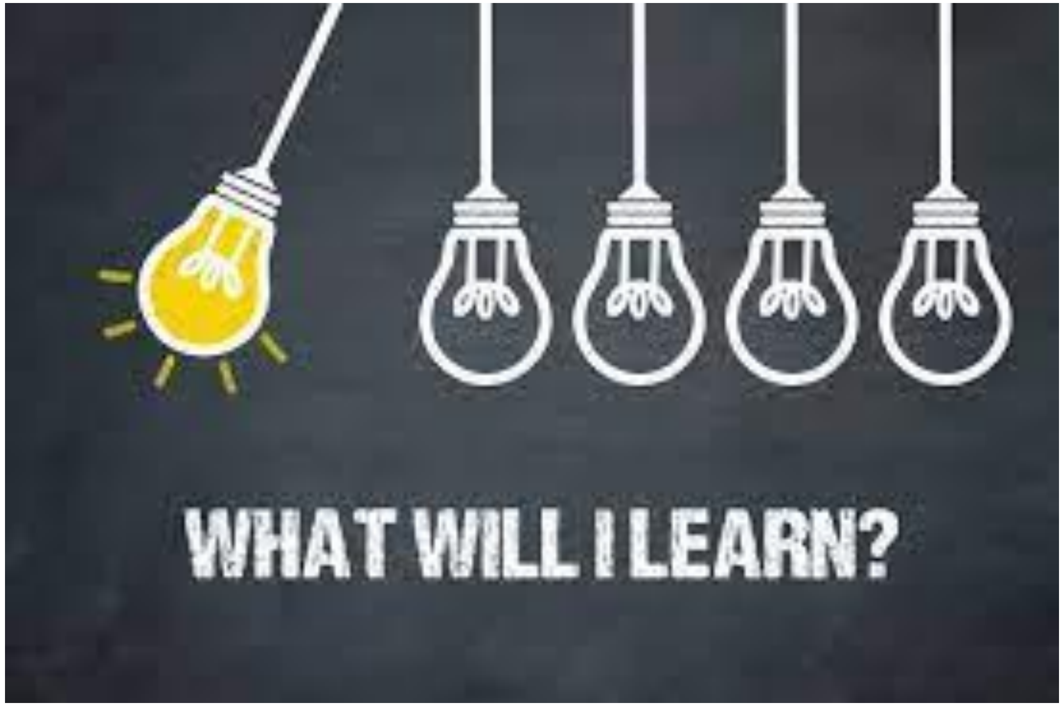
Before new line

Agenda Course 2

- Demand forecasting
  - Why forecasting demand
  - How to forecast demand
  - How to measure the quality of the forecast

  + Seminar o9 Guest speaker on ML and AI in forecasting
- Aggregate Planning
  - Planning Horizons
  - Aggregate Planning Problem
  - Unicorn case study

Learning Objectives



Explain the importance of forecasting

Review the different forecasting methods

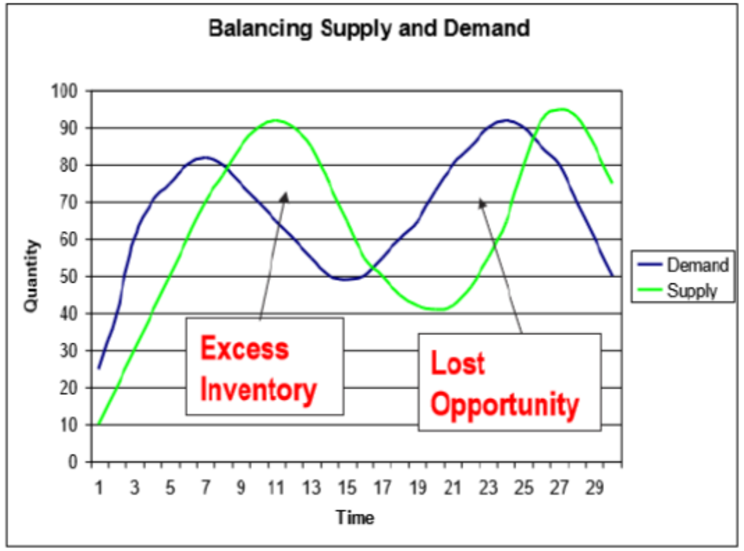
Apply the naive, moving average methods and regression analysis

Compute 3 KPI gauging the quality of the forecast

Learn latest forecasting techniques

Forecasting demand: why?

- Considering the often heard statement : 'forecast is always wrong' why do we spend a class on the topic?
- What is a forecast: an observation of data, enabling us to describe a future situation and taking action to face it at best we can
- True reality will turn differently but fail to plan means also plan to fail
- A forecast helps us to:
  - Plan ressources (human, equipment, inventory)
  - Align with our suppliers for raw materials
  - Satisfy customers
  - Communicate to investors our revenue plans
  - Avoid losing market shares



Impact of an inaccurate forecast

A Year Later, the Same Scene: Long Lines for the Elusive Wii

By MATT RICHTEL DEC. 14, 2007

"**Unmet demand is costing Nintendo**. Analysts estimate was that it lost 1 billiard USD during Xmas season , without counting the revenue from the games"

Hundreds of people hoping to buy a Nintendo Wii lined up Thursday at the Nintendo World store in Manhattan. Michael Nagle for The New York Times

Amazon Kindle Touch goes out of stock, sparks conspiracy theories

http://www.nytimes.com/2007/12/14/technology/14wii.html?_r=1&adxnnl=1&oref=slogin&adxnnlx=1197986742-oEOTYZw943dAye+uGcdfrg

https://www.engadget.com/2012/08/27/amazon-kindle-touch-goes-out-of-stock/

https://gigaom.com/2012/10/03/kindle-paperwhite-sold-out-until-november/

Forecast characteristics

- Expecting 100% accuracy is an illusion
- The more aggregated (product family over SKU) the better the accuracy
- Short-term forecast is more accurate than long-term forecast
- The more inputs you gather the better the accuracy
- Monitoring forecast accuracy helps to improve the situation
- Forecast needs interpretation; the human is in control of the decision of the final figure

The Value Proposition: What is the value of improving forecast accuracy?

Forecasting scenario		Value loss	Value leakage examples*
Imperfect demand signal	Under-forecasting	Revenue ↓	• Lost sales due to out-of-stocks and poor service levels / OTIF • Using lagging indicators of demand, no future drivers (e.g., trade, promos, events, etc.) • Lack ability to sense & respond to short-term demand requests (e.g., upside from events) • Lost sales due to poor New Product forecasts, hit rates and transition planning
		COGS ↑	• Increased production costs to fill and expedite backorders (e.g., overtime labor costs, expedited raw materials and packaging, etc.)
		Distribution Costs ↑	• Increased transportation costs from inaccurate forecasts at DC/location level (e.g., rebalancing, transfers, expedites, smaller volumes & more shipments, etc.)
	Over-forecasting	Revenue ↓	• Discounted sales due to disposal of slow moving / excess inventory • Using lagging indicators of demand, no future drivers (e.g., trade, promos, events, etc.) • Volatile demand due to portfolio fragmentation, diversity of distribution channels (incl. strong online growth) and competitor activity
		Distribution Costs ↑	• Increased transportation costs from inaccurate forecasts at DC/location level (e.g., rebalancing, transfers, expedites, more shipments and costs for disposals, etc..)
		Write-Off Costs ↑	• Additional costs to dispose and write-off slow-moving and obsolete inventory
		Inventory Carrying Costs ↑	• Additional costs for holding excess inventory at warehouses and off-site storage locations
		Working Capital ↑	• Excess safety stock and slow-moving inventory due to poor forecast accuracy and service levels – excess inventory in network to cover volatility

*Not-exhaustive, common examples of value leakage in CPG Industry

Improving forecast accuracy (reducing error) has direct impact on top and bottom-line performance.

Value targets for improved demand management

2x – 3x planner efficiency gains (touchless planning)

0.2% – 0.5% uplift in revenue with better SL, reduced OOS

5% – 15% reduction in inventory & holding cost

0.5% – 2%+ logistics costs reduction (expidited freight, labor & transfers)

10% – 20%+ obsolescence reduction

**Ranges are based on our experience at similar CPG companies who have achieved 5-10% improvement in forecast accuracy

How is this forecast used?

	Marketing	*Sales*	*Finance/ Accounting*	*Production/ Purchasing: Long Term*	*Production/ Purchasing: Short Term*	*Logistics: Long Term*	*Logistics: Short Term*
Needs	Annual plans (updated monthly or quarterly) for new and existing products or product changes, promotional efforts, channel placement and pricing	Setting goals for the sales force and motivating salespeople to exceed those goals	Projecting cost and profit levels and capital needs	Planning the development of plant and equipment	Planning specific production runs	Planning the development of storage facilities and transporation equipment	Specific decisions of what products to move to what locations and when
Level	Product or product line	Territory and/or customers	Corporate, division, product line	Product (SKU)	Product (SKU)	Product by location (SKUL)	Product by location (SKUL)
Horizon	Annual	1-2 years	1-5 years	1-3 years	1-6 months	Monthly to several years	Daily, weekly, monthly
Interval	Monthly or quarterly	Monthly or quarterly	Monthly or quarterly	Quarterly	Daily, weekly, monthly	Monthly	Daily, weekly, monthly
Form	Dollars	Dollars	Dollars	Units	Units	Units/Weight/Cube	Units/Weight/Cube

Forecasting 7 steps process

1. Determine the use of the forecast
2. Select the item to be forecasted
3. Determine the horizon
4. Select the model(s)
5. Gather data
6. Make the forecast
7. Validate the results and implement

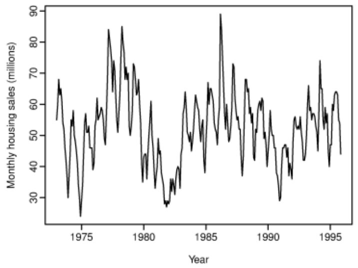
Forecasting Techniques

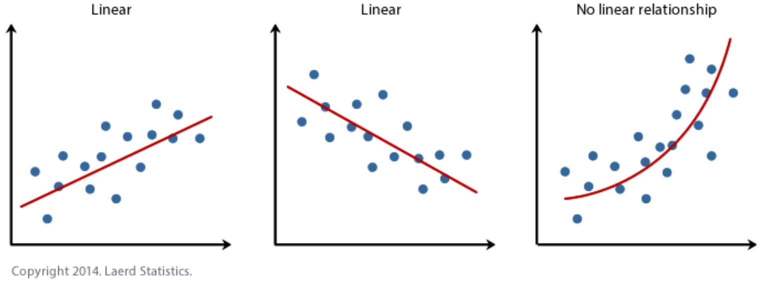
- Quantitative and qualitative approaches
- Within the quantitative approaches there is a large variety of techniques and it might sound confusing
- Different techniques work best in different situations.

Time serie
Open model
Fixed model

Regression

Judgmental
Delphi method
Customer survey

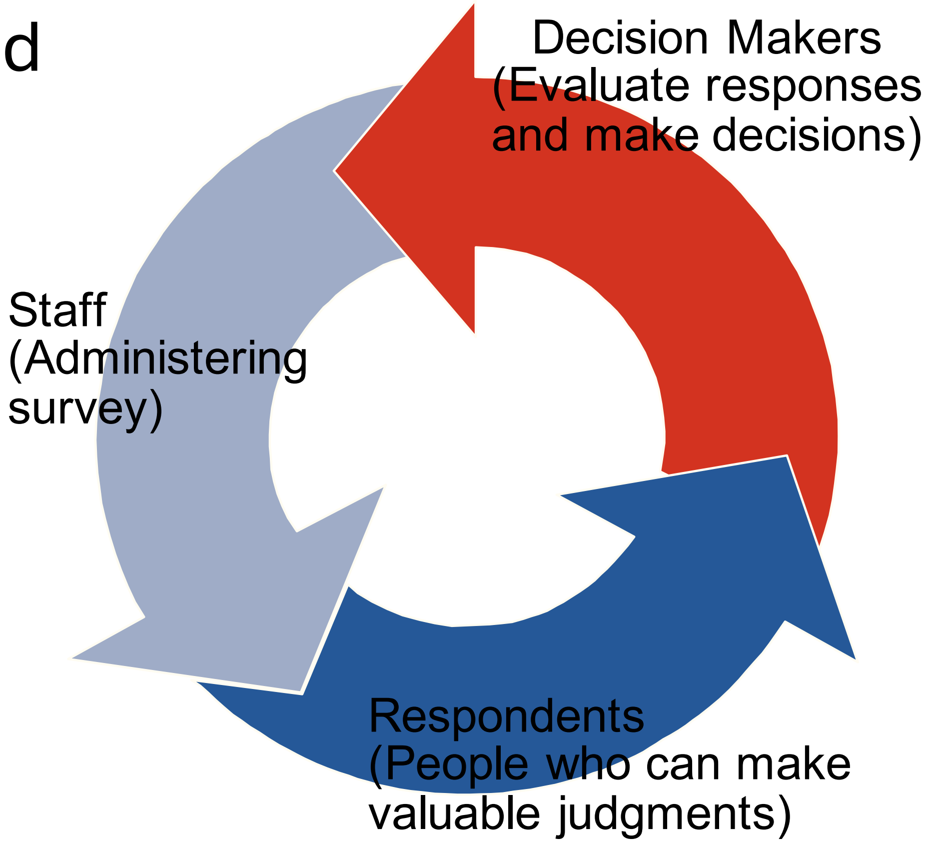




- Quantitative techniques support a better decision BUT are more costly and are not taking the decision in the end!!!

Judgemental: Delphi method

- Iterative process involving experts in the field
- Advantages:
  - Consensus of experts
  - Effective for long-term and new products
- Drawbacks:
  - Lengthy
  - Expensive
  - Questionable reliability



Heizer, Render and Munson 2020
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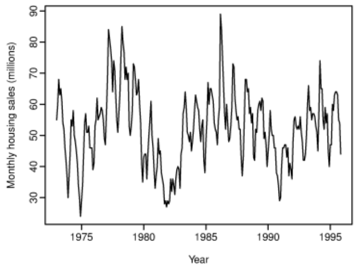
Forecasting Techniques

Time serie
Open model
Fixed model



Time-serie:

The past explains the future

- **Naïve model**: simple and responsive: Forecast at t equal actual at t-1

$$\boldsymbol{F_t} = \boldsymbol{A_{t-1}}$$

- **Cumulative model** (weighted, exponentially smoothed or not): simple and more stable

$$\text{Moving } average = \frac{\sum \text{demand in previous } n \text{ periods}}{n}$$

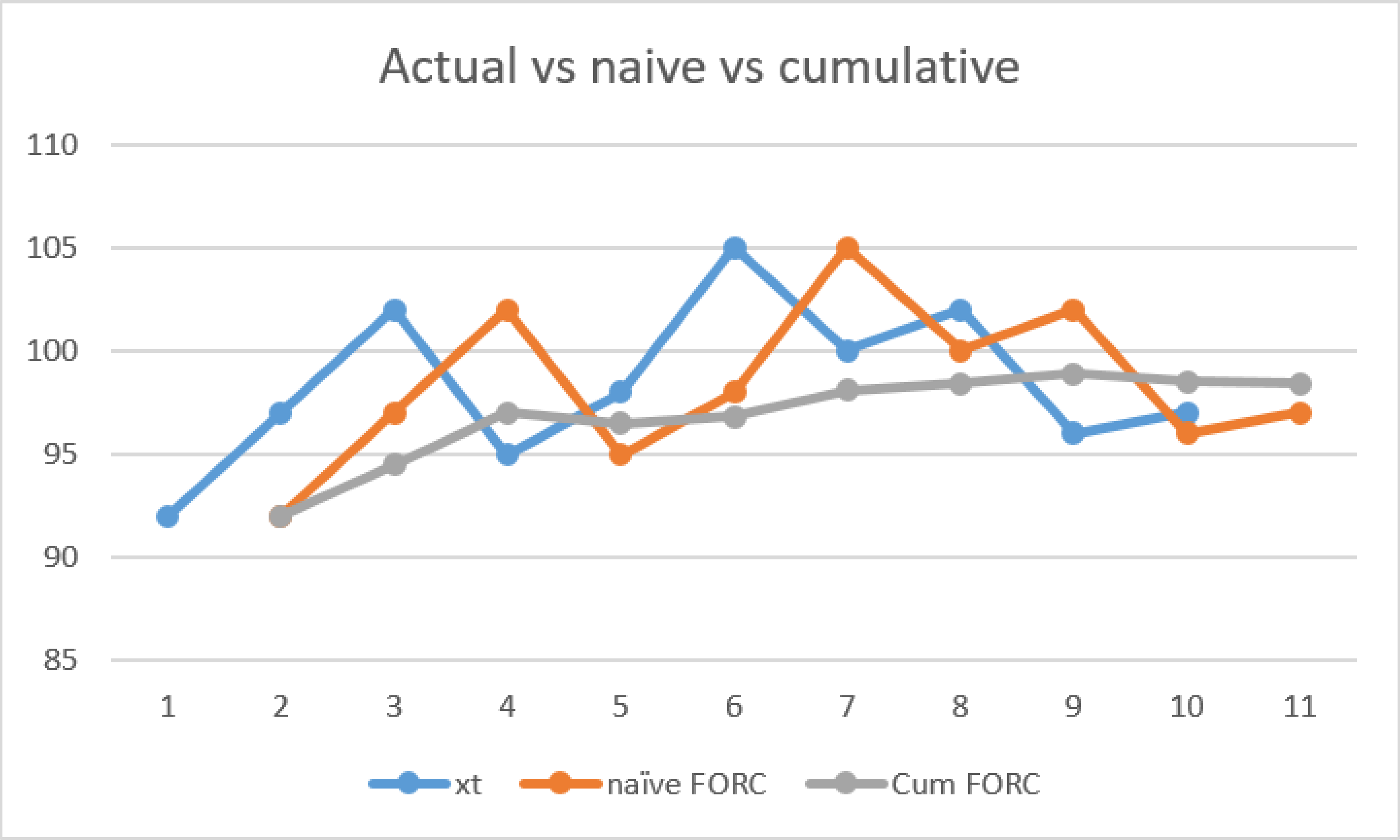
$$Weigthed\ moving\ average = \frac{\sum[(\text{Weight for period } n) \times (\text{Demand in period } n)]}{\sum \text{Weights}}$$

$$Exponetially\ smooth\ average\ F_t = F_{t-1} + \alpha(A_{t-1} - F_{t-1})$$

α : smoothing constant between 0 and 1, decided by the analyst

Time-serie

t	xt	naïve FORC	Cum FORC
1	92		
2	97	92	92
3	102	97	94.5
4	95	102	97
5	98	95	96.5
6	105	98	96.8
7	100	105	98.1
8	102	100	98.4
9	96	102	98.9
10	97	96	98.5
11		97	98.4



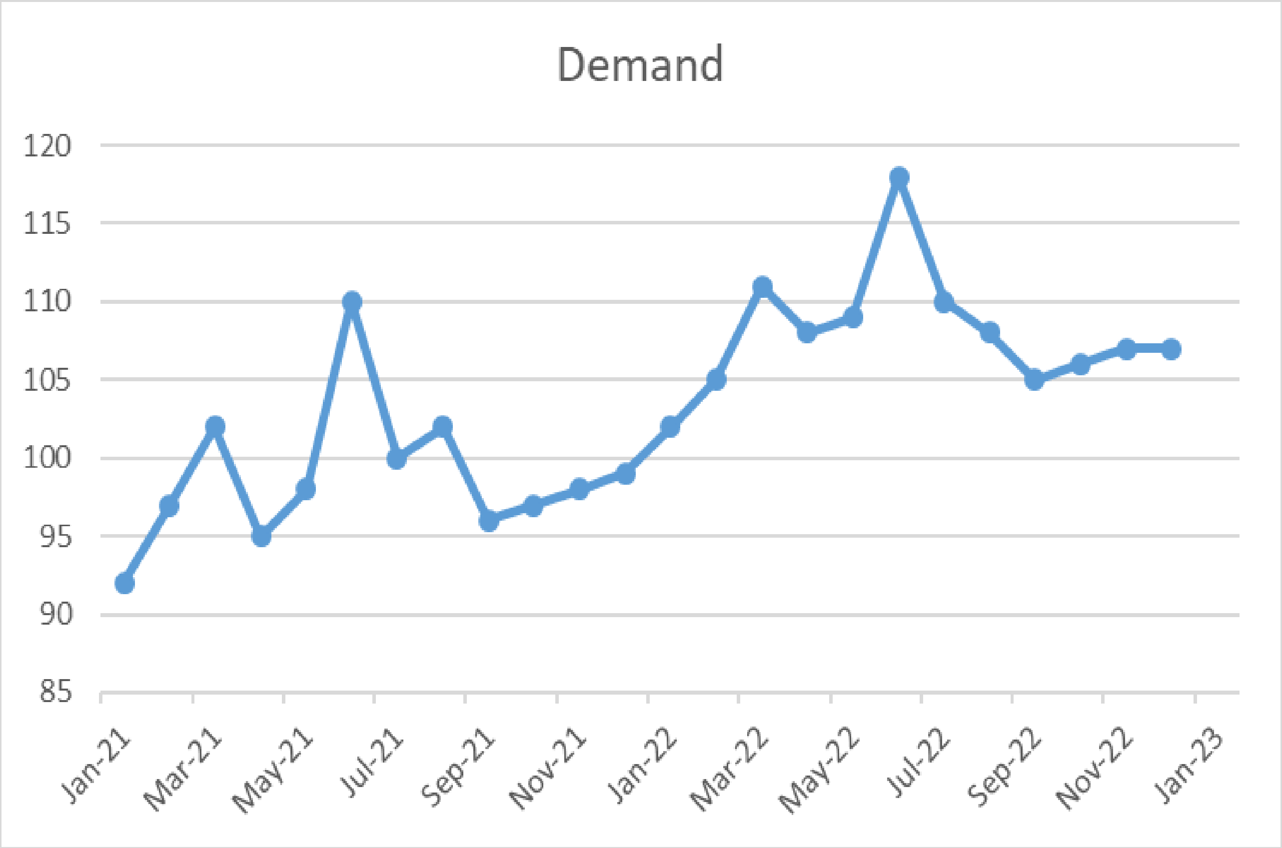
The naïve model is very responsive and would work well with changing conditions but might bring instabilities in the downstream process.

The cumulative model is exactly the opposite.

Time serie forecasting

- Question 1 : look at the data. Do you detect any trend over the last 2 years ?
  - Yes positive
  - No demand is stationary
  - Yes negative
  - Cannot say
- Question 2 : Is there a seasonality
  - No
  - Not enough data to say
  - Yes of sort
- What would be the forecast for Jan 2023:
  - A naïve model
  - A cumulative model over 4 months

Fixed Model time serie: level models case study



t	Demand
Jan-21	92
Feb-21	97
Mar-21	102
Apr-21	95
May-21	98
Jun-21	110
Jul-21	100
Aug-21	102
Sep-21	96
Oct-21	97
Nov-21	98
Dec-21	99
Jan-22	102
Feb-22	105
Mar-22	111
Apr-22	108
May-22	109
Jun-22	118
Jul-22	110
Aug-22	108
Sep-22	105
Oct-22	106
Nov-22	107
Dec-22	107
Jan-23	

Time-serie: exponentially smooth average

- The smoothing constant (α) is used for the weighting
  - $0 \leq \alpha \leq 1$, but usually, $0{,}05 \leq \alpha \leq 0{,}50$
  - A large α gives more weight to recent data (used when the average is changing)
  - A smaller α gives more weight to old data(used when the underlying average is stable)

$$Exponetially\ smooth\ average\ F_t = F_{t-1} + \alpha(A_{t-1} - F_{t-1})$$

α : smoothing constant between 0 and 1, decided by the analyst

Time Series Forecasting: exponential smoothing Example 1

Sales of Volkswagen's popular Beetle have increased steadily at car dealerships over the past 5 years. The sales manager had predicted before the introduction of the new model that first-year sales would be 410 VW. Using exponential smoothing with a weight of α=0.30, develop forecasts for years 2 to 6.

YEAR	SALES	FORECAST
1	450	410
2	495	
3	518	
4	563	
5	584	
6	?	

$$F_t = F_{t-1} + \alpha(A_{t-1} - F_{t-1})$$

Time Series Forecasting: exponential smoothing Example 2

Sales of a brand new smart watch are expecting to be outstanding. The sales manager had predicted before the introduction of the new model that first-year sales would be 100. Using exponential smoothing with a weight of α=0.40, develop forecasts for years 2 to 5.

What do you observe?

YEAR	SALES	FORECAST
1	100	100
2	200	
3	300	
4	400	
5	500	

$$F_t = F_{t-1} + \alpha(A_{t-1} - F_{t-1})$$

Time-serie

Method	Advantages	Drawbacks	Best Used For...
Naïve	Simple to use; zero cost; no data history required.	Highly reactive; ignores trends or seasonality; usually inaccurate.	Stable products with no growth or decline.
Moving Average (MA)	Smooths out "noise" (random spikes); easy to calculate.	Always lags behind actual trends; ignores older data that might be relevant.	Products with stable demand and low volatility.
Weighted Moving Average (WMA)	More responsive than MA; allows you to prioritize recent sales data.	Harder to determine the "correct" weights; still lags behind rapid shifts.	Products where the most recent month is a strong indicator of the next.
Exponential Smoothing (ES)	Requires very little data storage; highly flexible via the smoothing constant (α).	Can be complex to "tune" the α value; doesn't handle strong seasonality well without adjustment.	Broad range of inventory, especially when trends change quickly.

Time-serie: exponential smoothing with trend

Forecast including trend	(FIT_t) =	Exponentially smoothed forecast	(F_t)	+	Exponentially smoothed trend	(T_t)

Time-serie: exponential smoothing with trend

Steps

1. Calculate F_t , the exponentially smoothed forecast for period t (Eq2)
2. Calculate the exponentially smoothed trend $\mathrm{T_t}$ (Eq3)
3. Calculate the forecast, FIT_t, with $FIT_t = F_t + T_t$ (Eq1)

$$\text{(Eq1) } FIT_t = F_t + T_t$$

$$\text{(Eq2) } F_t = \alpha(A_{t-1}) + (1-\alpha)(F_{t-1} + T_{t-1})$$

$$\text{(Eq3) } T_t = \beta(F_t - F_{t-1}) + (1-\beta)T_{t-1}$$

where,
F_t = exponentially smoothed forecast average in period t
T_t = exponentially smoothed trend in period t
A_t = actual demand in period t
α = smoothing (or weighting) constant for the average, $0 \le \alpha \le 1$
β = smoothing (or weighting) constant for the average, $0 \le \beta \le 1$

Time-serie: exponential smoothing with trend Example

MONTH (t)	ACTUAL DEMAND (A_t)	MONTH (t)	ACTUAL DEMAND (A_t)
1	12	6	21
2	17	7	31
3	20	8	28
4	19	9	36
5	24	10	?

$\alpha = .2$ $\beta = .4$ $T_1 = 2$ units $F_1 = 11$ units

Ch04, Example 7, Heizer, Render and Munson 2020

Time-serie: exponential smoothing with trend

Example $F_t = \alpha(A_{t-1}) + (1-\alpha)(F_{t-1} + T_{t-1})$ $T_t = \beta(F_t - F_{t-1}) + (1-\beta)T_{t-1}$

TABLE 4.2 **Forecast with $\alpha = .2$ and $\beta = .4$**

MONTH	ACTUAL DEMAND	SMOOTHED FORECAST AVERAGE, F_t	SMOOTHED TREND, T_t	FORECAST INCLUDING TREND, FIT_t
1	12	11	2	13.00
2	17			
3	20			
4	19			
5	24			
6	21			
7	31			
8	28			
9	36			
10	—			

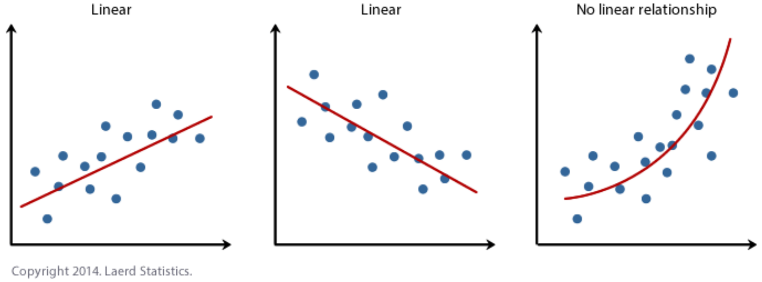
Time-serie: exponential smoothing with trend

Example

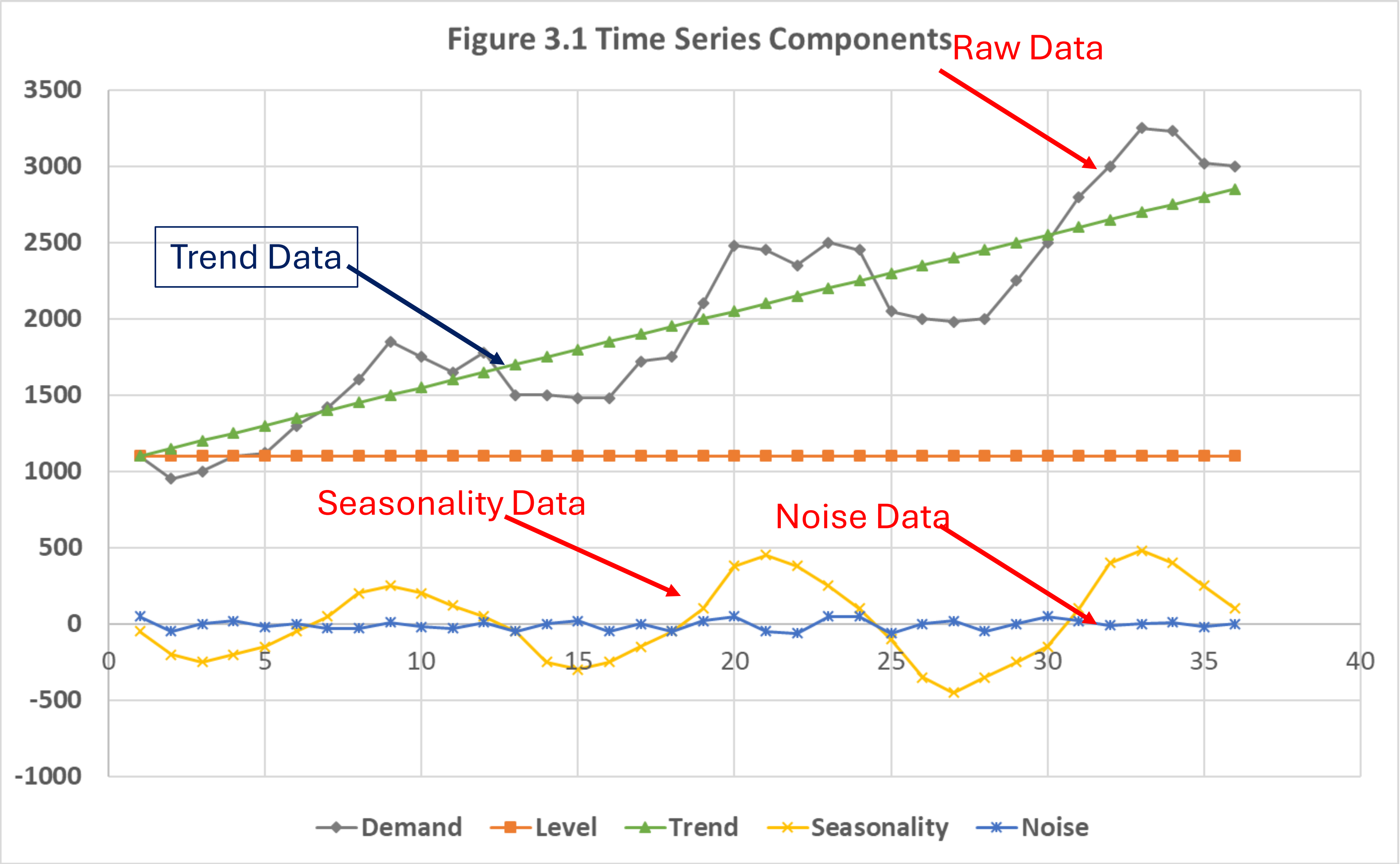
TABLE 4.2	Forecast with $\alpha = .2$ and $\beta = .4$					
MONTH	ACTUAL DEMAND	SMOOTHED FORECAST AVERAGE, F_t		SMOOTHED TREND, T_t	FORECAST INCLUDING TREND, FIT_t	
1	12	11	Error	2	13.00	Error
2	17	12.8	24%	1.92	14.72	13%
3	20	15.18		2.10	17.28	
4	19	**17.82**		**2.32**	**20.14**	
5	24	**19.91**	17%	**2.23**	**22.14**	7,8%
6	21	**22.51**		**2.38**	**24.89**	
7	31	**24.11**		**2.07**	**26.18**	
8	28	**27.14**	3%	**2.45**	**29.59**	5%
9	36	**29.28**		**2.32**	**31.60**	
10	—	**32.48**		**2.68**	**35.16**	

Forecasting Techniques

Regression



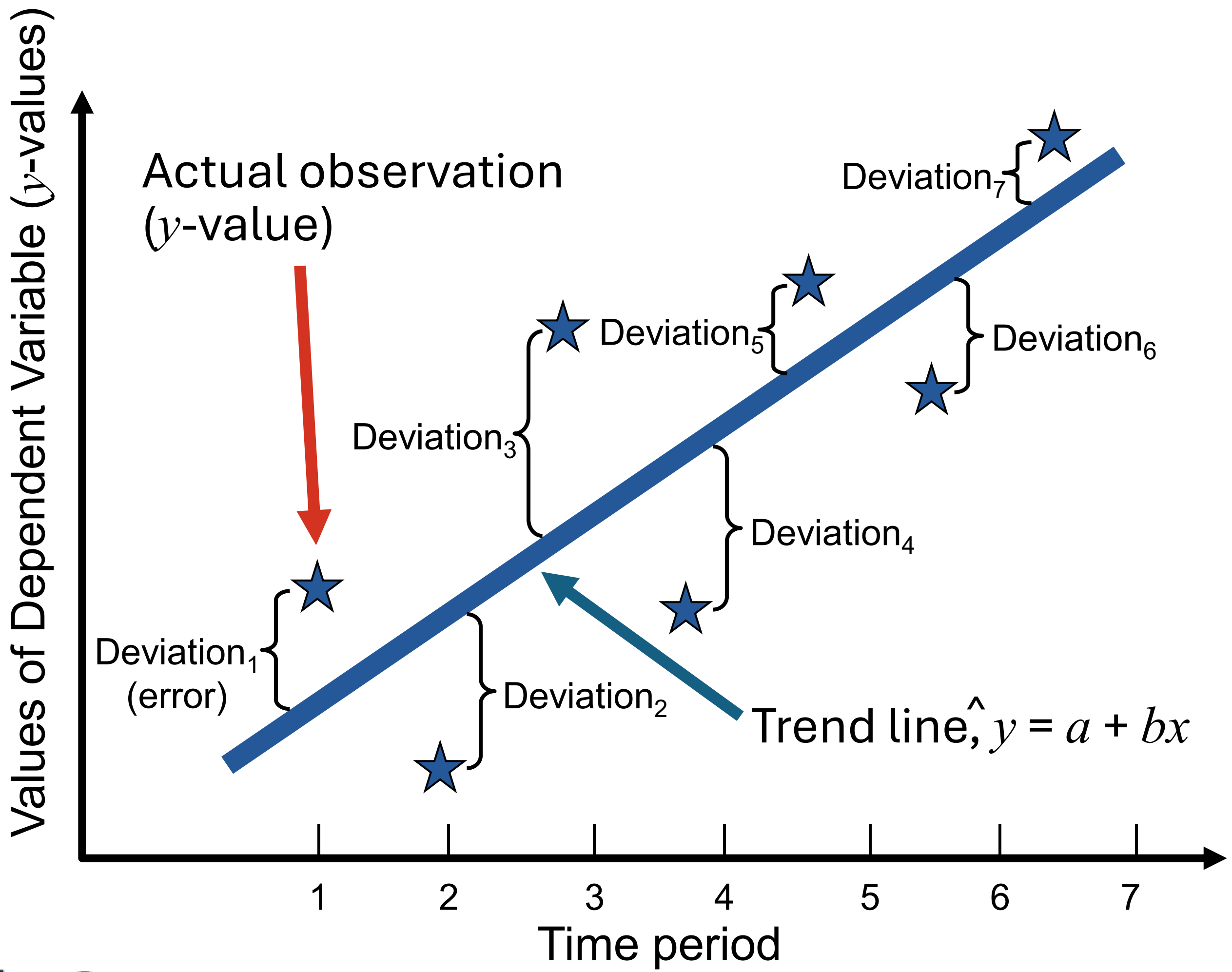
Closer to reality



1. Trend
   - Up/downward movement of data along the time dimension
2. Seasonality
   - The demand varies upon quarters or months in a regular way
3. Noise
   - "Blips" in the data due to random effects

Linear Regression analysis

- Statistical method
- The technique tries to establish a relationship between external data (like advertising) or time and the demand.
- If a correlation is found the external data can be used to forecast future demand
- This is the most accurate technique but requires a large amount of data.
- It can be slow and might miss short-term changes in the trend.



Ch04, Fig. 4.4, Heizer, Render and Munson 2020

- Curve Fitting . Best Fit: Simple Linear Regression
  - One Independent Variable (X) is used to predict One Dependent Variable (Y)
  - $Y = a + b X$

Linear Regression analysis

$$\hat{y} = a + bx$$

where, $\hat{y} = \textit{calculated value of the data we are trying to predict}$

$a = y \text{ intercept}$

$b = slope$

$x = \text{independant variable}$

$$\boxed{b = \frac{\sum xy - n\bar{x}\bar{y}}{\sum x^2 - n\bar{x}^2}} \qquad \boxed{a = \bar{y} - b\bar{x}}$$

These are the **formulas** to solve for coefficients *a* and *b*. These only work for linear regressions **with 1 *x* variable.**

where, $x = \text{known values of the independent variable}$

$y = \text{known values of the dependent variable}$

$\bar{x} = \text{average of the } x \text{ values}$

$\bar{y} = \text{average of the of the } y \text{ values}$

$n = \text{the number of data points or observation}$

Linear Regression analysis: standard deviation

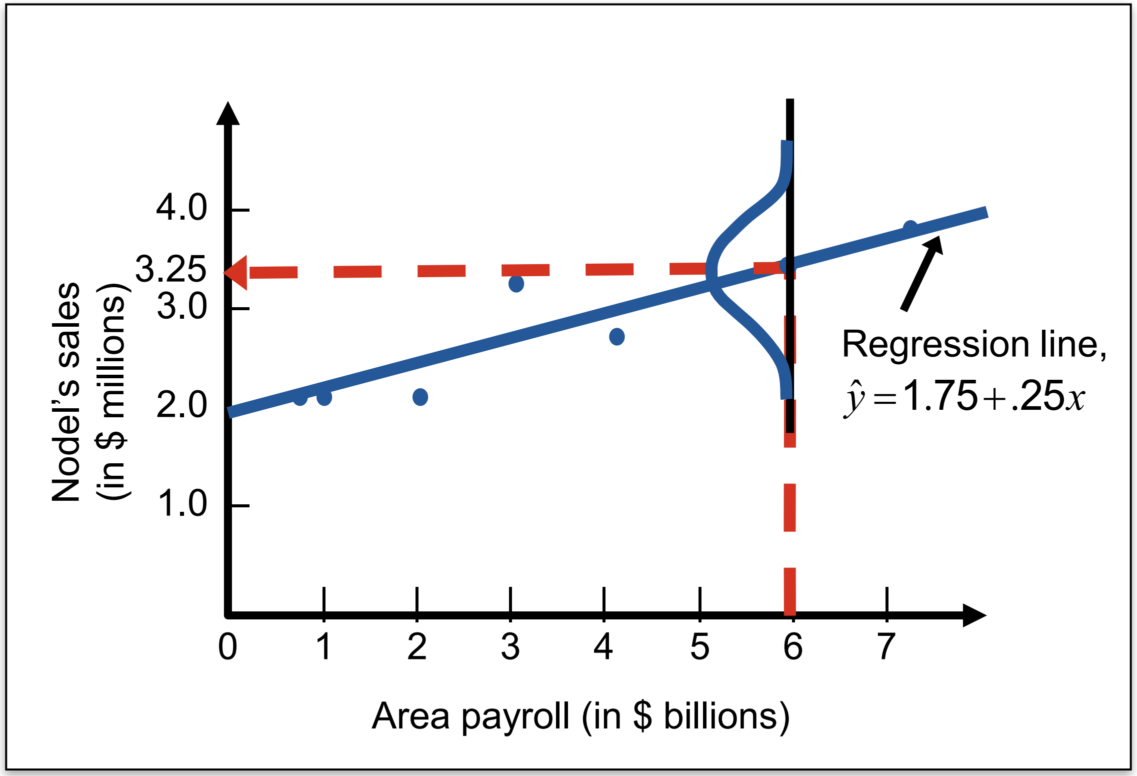
How far is our estimated value to the real value?

$$S_{y,x} = \sqrt{\frac{\sum(y - y_c)^2}{n-2}} = \sqrt{\frac{\sum y^2 - a\sum y - b\sum xy}{n-2}}$$

where, y = y value of each data point

y_c = computed value of the dependent variable from the regression equation

n = the number of data points or observation



Ch04, Fig. 4.9, Heizer, Render and Munson 2020

Linear Regression analysis:

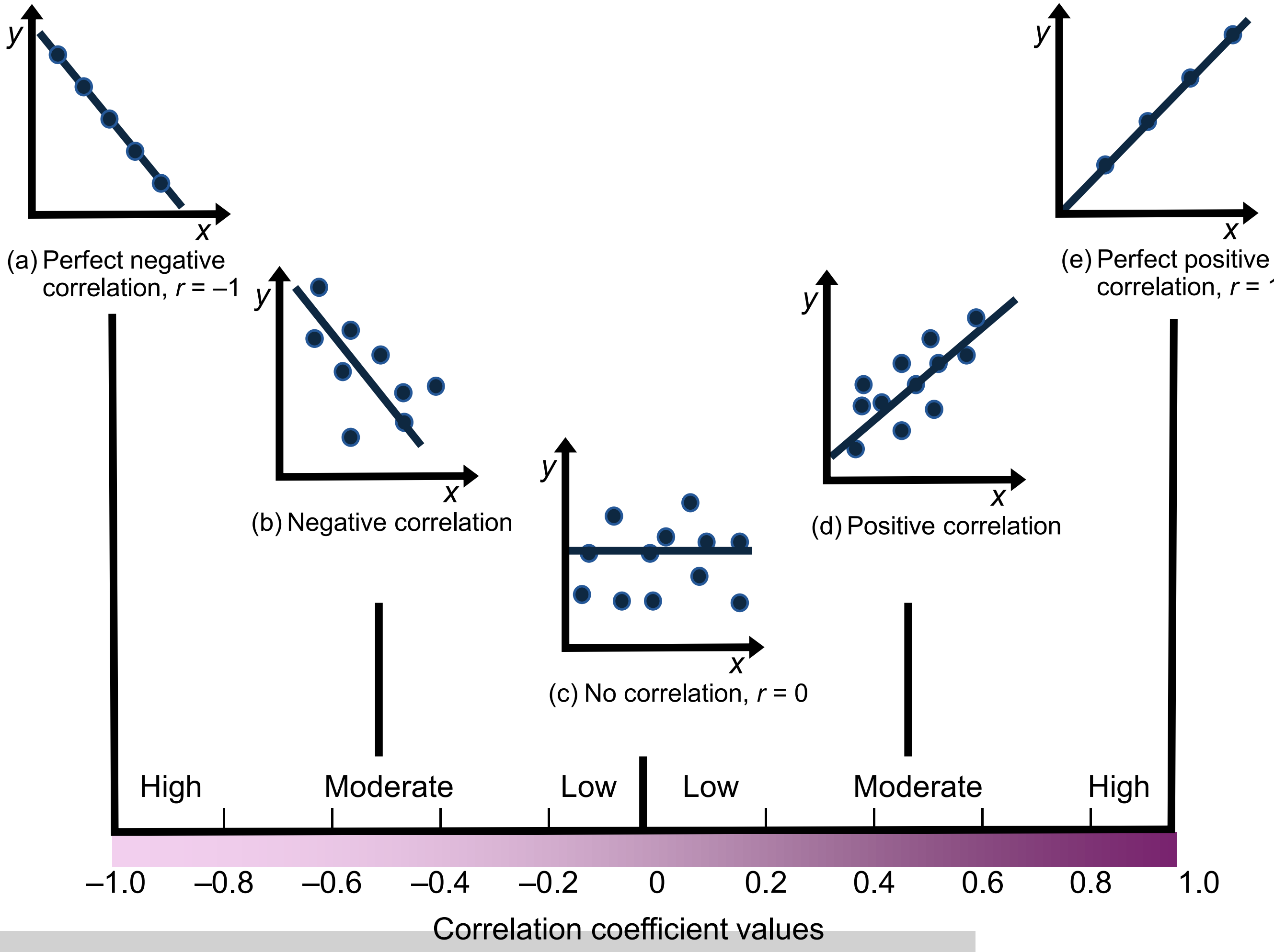
correlation coefficient and **determination** coefficient

r shows how strong is the relationship between x and y ie is x usefull to predict y

r2 shows how well the model predict the dependant variable

$$r = \frac{n\sum xy - \sum x \sum y}{\sqrt{[n\sum x^2 - (\sum x)^2][n\sum y^2 - (\sum \boldsymbol{y})^2]}}$$

r is between -1 to 1



Ch04, Fig. 4.10, Heizer, Render and Munson 2020

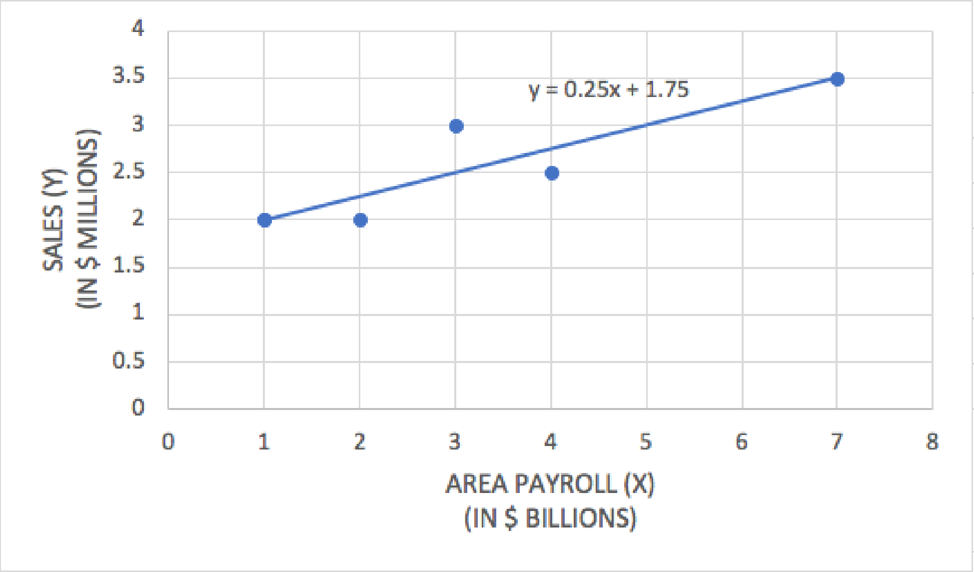
Example

- Nodel Construction renovates old house. Over the years it found out that its business was linked to the payroll in its neighborhood.
- They want to build a model to predict future business

AREA PAYROLL (X) (IN $ BILLIONS)	SALES (Y) (IN $ MILLIONS)
1	2
3	3
4	2.5
2	2
1	2
7	3.5

Example

AREA PAYROLL (X) (IN $ BILLIONS)	SALES (Y) (IN $ MILLIONS)	X^2	XY	Y^2
1	2	1	2	4
3	3	9	9	9
4	2.5	16	10	6.25
2	2	4	4	4
1	2	1	2	4
7	3.5	49	24.5	12.25
$\sum x =$ 18	$\sum y =$ 15	$\sum x^2 =$ 80	$\sum xy =$ 51.5	$\sum y^2 =$ 39.5
$\bar{x} =$ 3	$\bar{y} =$ 2.50			



$$b = \frac{\sum xy - n\bar{x}\bar{y}}{\sum x^2 - n\bar{x}^2} \qquad a = \bar{y} - b\bar{x} \qquad S_{y,x} = \sqrt{\frac{\sum y^2 - a\sum y - b\sum xy}{n-2}} \qquad r = \frac{n\sum xy - \sum x \sum y}{\sqrt{[n\sum x^2 - (\sum x)^2][n\sum y^2 - (\sum y)^2]}}$$

	Coefficients
b	0.250
a	1.750

$S_{y,x}$	0.306
r	0.901
r^2	0.81

Example: interpretation

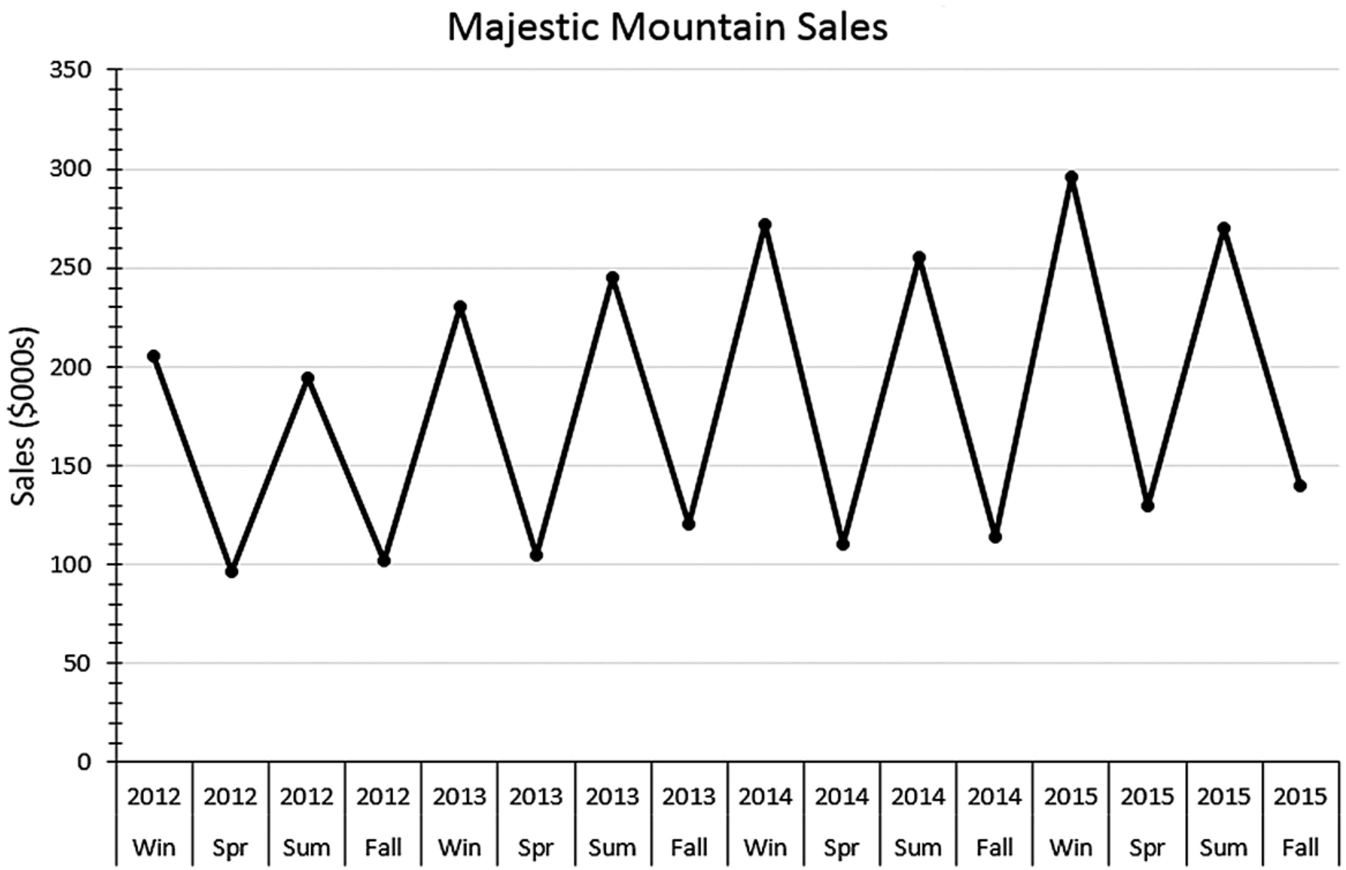
- How to interpret the results?
- Next month the local chamber of commerce predicts payroll to be 6 millions USD, how is Nodel business going to be? 3,25 millions USD
- What is the potential error? Sxy = 306000 USD
- What their observation of a link accurate? Yes r=0,91
- r2=0,81: 81% of the total variation is explained by the model

Forecasting Techniques

Seasonality

Adjustments for seasonality

The owners of the ski resort Majestic Mountain want to develop a model for their quarterly sales. They have the 2012 to 2015 data on hand. There is a seasonal pattern with peaks in summer and winters.



Adjustments for seasonality

Isolation and evaluation of seasonal indexes:

Multiplicative Time-Series Model

$$y_t = T_t \times S_t \times C_t \times I_t$$

where:

y_t = Value of the time series at time t
T_t = Trend value at time t
S_t = Seasonal value at time t
C_t = Cyclical value at time t
I_t = Irregular or random value at time t

With our method we will assume the effect of I and C are negligeable.

Adjustments for seasonality

Calculate the moving average over 4 periods (when you have quarterly data)

	A	B	C	D	E
1	Season	Year	Quarter	Sales	4-Period Moving Average
2	Winter	2012	1	205	
3	Spring		2	96	149.25
4	Summer		3	194	155.50
5	Fall		4	102	157.75
6	Winter	2013	5	230	170.50
7	Spring		6	105	175.00
8	Summer		7	245	185.50
9	Fall		8	120	186.75
10	Winter	2014	9	272	189.25
11	Spring		10	110	187.75
12	Summer		11	255	193.75
13	Fall		12	114	198.75
14	Winter	2015	13	296	202.50
15	Spring		14	130	209.00
16	Summer		15	270	
17	Fall		16	140	
18					

Each average smooth out the yearly data and try to soften the seasonality effect.

Adjustments for seasonality

	A	B	C	D	E	F
1	Season	Year	Quarter	Sales	4-Period Moving Average	Centered Moving Average
2	Winter	2012	1	205		
3	Spring		2	96	149.25	
4	Summer		3	194	155.50	152.375
5	Fall		4	102	157.75	156.625
6	Winter	2013	5	230	170.50	164.125
7	Spring		6	105	175.00	172.750
8	Summer		7	245	185.50	180.250
9	Fall		8	120	186.75	186.125
10	Winter	2014	9	272	189.25	188.000
11	Spring		10	110	187.75	188.500
12	Summer		11	255	193.75	190.750
13	Fall		12	114	198.75	196.250
14	Winter	2015	13	296	202.50	200.625
15	Spring		14	130	209.00	205.750
16	Summer		15	270		
17	Fall		16	140		

$$T_t \text{ x } C_t = \textit{Centered MA}$$

$$\textit{Centered MA} = \frac{149.25 + 155.50}{2}$$

$$\textit{Centered MA} = 152.375$$

The centered MA allocates a more centered value to the quarter we are interested in.

Adjustments for seasonality

Ratio-to-Moving-Average

$$S_t \times I_t = \frac{y_t}{T_t \times C_t}$$

	A	B	C	D	E	F	G
1	Season	Year	Quarter	Sales	4-Period Moving Average	Centered Moving Average	Ratio to Moving Average
2	Winter	2012	1	205			
3	Spring		2	96	149.25		
4	Summer		3	194	155.50	152.38	1.27
5	Fall		4	102	157.75	156.63	0.65
6	Winter	2013	5	230	170.50	164.13	1.40
7	Spring		6	105	175.00	172.75	0.61
8	Summer		7	245	185.50	180.25	1.36
9	Fall		8	120	186.75	186.13	0.64
10	Winter	2014	9	272	189.25	188.00	1.45
11	Spring		10	110	187.75	188.50	0.58
12	Summer		11	255	193.75	190.75	1.34
13	Fall		12	114	198.75	196.25	0.58
14	Winter	2015	13	296	202.50	200.63	1.48
15	Spring		14	130	209.00	205.75	0.63
16	Summer		15	270			
17	Fall		16	140			
18							

$$S_3 \text{ x } I_3 = \frac{y_3}{T_3 \text{ x } C_3}$$

$$S_3 \text{ x } I_3 = \frac{194}{152.38} = 1.27$$

Rounded to 2 decimal places

Adjustments for seasonality

Calculation of seasonal indexes:

Summer: $$\frac{1.27+1.36+1.34}{3}=1.323$$

Seasonal Indexes

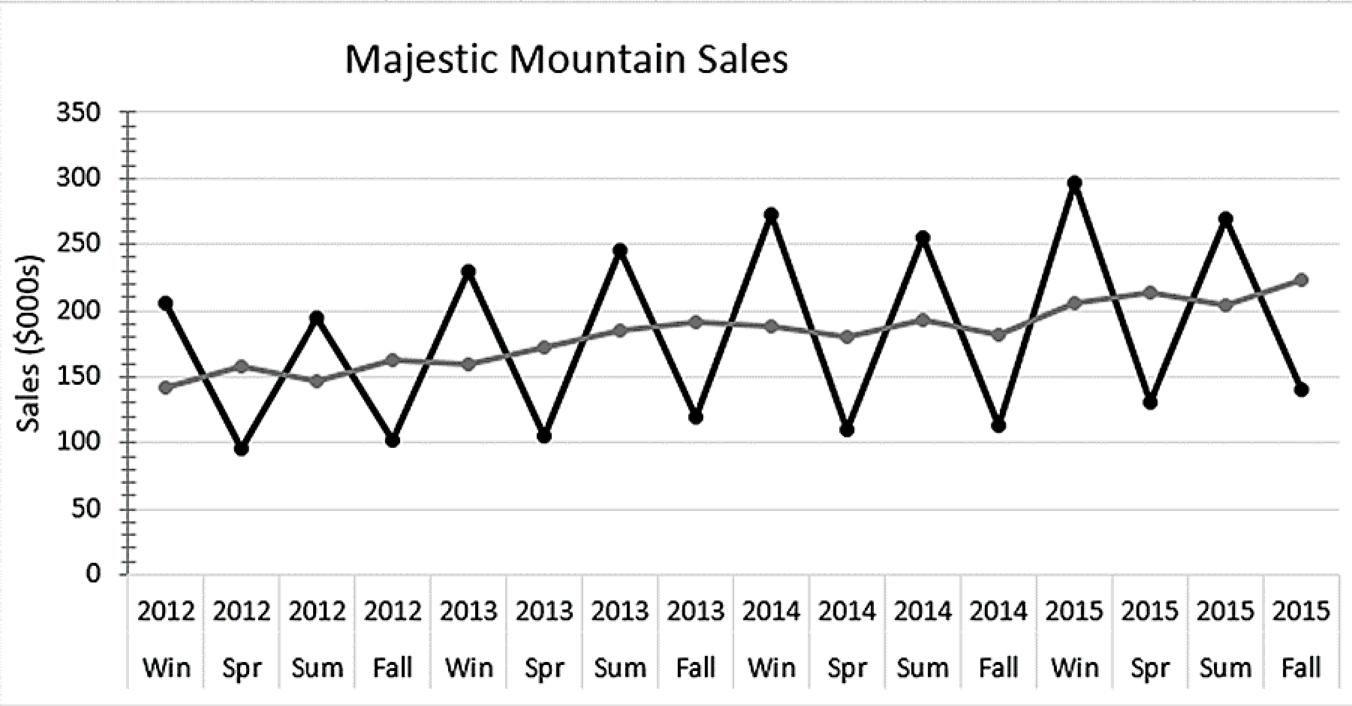
20	Summer	Fall	Winter	Spring	
21	1.273	0.651	1.401	0.608	
22	1.359	0.645	1.447	0.584	
23	1.337	0.581	1.475	0.632	
24					
25	3.969	1.877	4.324	1.823	= Total
26	1.323	0.626	1.441	0.608	= Average

Adjustments for seasonality

Deseasonalization

$$T_t \times C_t \times I_t = \frac{y_t}{S_t}$$

	A	B	C	D	E	F	G	H	I
1	Season	Year	Quarter	Sales	4-Period Moving Average	Centered Moving Average	Ratio to Moving Average	Seasonal Index	Deseasonalized Data
2	Win	2012	1	205				1.441	142.26
3	Spr	2012	2	96	149.25			0.608	157.89
4	Sum	2012	3	194	155.50	152.38	1.273	1.323	146.64
5	Fall	2012	4	102	157.75	156.63	0.651	0.626	162.94
6	Win	2013	5	230	170.50	164.13	1.401	1.441	159.61
7	Spr	2013	6	105	175.00	172.75	0.608	0.608	172.70
8	Sum	2013	7	245	185.50	180.25	1.359	1.323	185.19
9	Fall	2013	8	120	186.75	186.13	0.645	0.626	191.69
10	Win	2014	9	272	189.25	188.00	1.447	1.441	188.76
11	Spr	2014	10	110	187.75	188.50	0.584	0.608	180.92
12	Sum	2014	11	255	193.75	190.75	1.337	1.323	192.74
13	Fall	2014	12	114	198.75	196.25	0.581	0.626	182.11
14	Win	2015	13	296	202.50	200.63	1.475	1.441	205.41
15	Spr	2015	14	130	209.00	205.75	0.632	0.608	213.82
16	Sum	2015	15	270				1.323	204.08
17	Fall	2015	16	140				0.626	223.64
18									



Adjustments for seasonality

Use the deseasonalized data to calculate the linear regression for the trend

SUMMARY OUTPUT

Regression Statistics	
Multiple R	0.9416
R Square	0.8865
Adjusted R Square	0.8784
Standard Error	8.2557
Observations	16

$$F_t = 142.096 + 4.683t$$

ANOVA

	df	*SS*	*MS*	*F*	*Significance F*
Regression	1	7455.934	7455.934	109.394	0.0000
Residual	14	954.191	68.156		
Total	15	8410.125			

	Coefficients	*Standard Error*	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*
Intercept	142.096	4.329	32.822	0.000	132.810	151.381
Quarter	4.683	0.448	10.459	0.000	3.723	5.643

Adjustments for seasonality

Calculate the forecast for quarters 17 à 20 (2016):

$$F_{17} = 142.096 + 4.683(17) = 221{,}707$$

Quarter (2015)	t	Unadjusted Forecast
Winter	17	221.707
Spring	18	226.390
Summer	19	231.073
Fall	20	235.756

Adjustments for seasonality

Reinject the seasonality in the data to get closer to reality

$$F_{t_{adjusted}} = F_{t_{unadjusted}} * Index$$

$$F_{17} = 221.707 * 1.441 = 319.48$$

Quarter (2015)	t	Unadjusted Forecast	Index	Adjusted Forecast
Winter	17	221.707	1.441	= 319,480
Spring	18	226.390	0.608	= 137,645
Summer	19	231.073	1.323	= 305,710
Fall	20	235.756	0.626	= 147,583

Forecasting Techniques

Future techniques

o9 supports companies along their forecasting maturity journey

Simple Time Series Forecasting

- Low complexity, out-of-box models
- Exponential smoothing, Linear, Moving average, Naive

Modest accuracy*

SKU-Level Statistical Forecasting

- Create forecast segments based on volume, criticality, variability, PLC, intermittency
- History cleansing outlier and missing value
- Best model selected from out-of-sample simulation
- Advanced methods including Exponential Smoothing, ARIMA, TBATS, STLM, GARCH, Crostons, SBA, State Space models, theta

Improved accuracy **

Multi-level Hierarchical Forecasting

- Intelligent Data Mining techniques to extract and identify Product and Location attributes
- Segmentation to detect key levels of forecast that aligns with business
- Ability to utilize seasonality, trend, cyclicity at multiple levels to enhance the planning/lowest level forecast
- Reconcile forecasts by overlaying hierarchical splits on rule based best-fit output
- Apply linear promotional lifts

High accuracy **

Driver Based ML Forecasting

- Advanced ML models based on historical feature engineering and product/store attributes
- Demand Sensing via promotions, pricing, competitor activity, holidays, local events, policy changes, macroeconomic factors, industry data, weather
- Advanced pattern recognition allows to accurately model slow-moving items and transition/NPIs
- Consider Inventory levels and backorders
- Auto bias correction & Exception handling
- Highly customizable model allows for business/data driven constraints
- Random Forest, GBM, Neural Networks

Class leading accuracy **

*Accuracy Computed as: 1 – wMAPE
**Results based on internal simulations. Actual results may vary.
***Input drivers depend on data availability

Demand forecast KPIs

- Why do we need KPIs ?
- To gauge how our demand planning process performs and drives continuous improvements.
- 3 broadly used KPIs:
  - Forecast accuracy (how much did we sell versus how much we forecasted for a given period)
  - Forecast consistency (how is our forecast changing month by month)
  - Forecast bias (do we have a tendency to over/under forecast)

Demand forecast KPIs

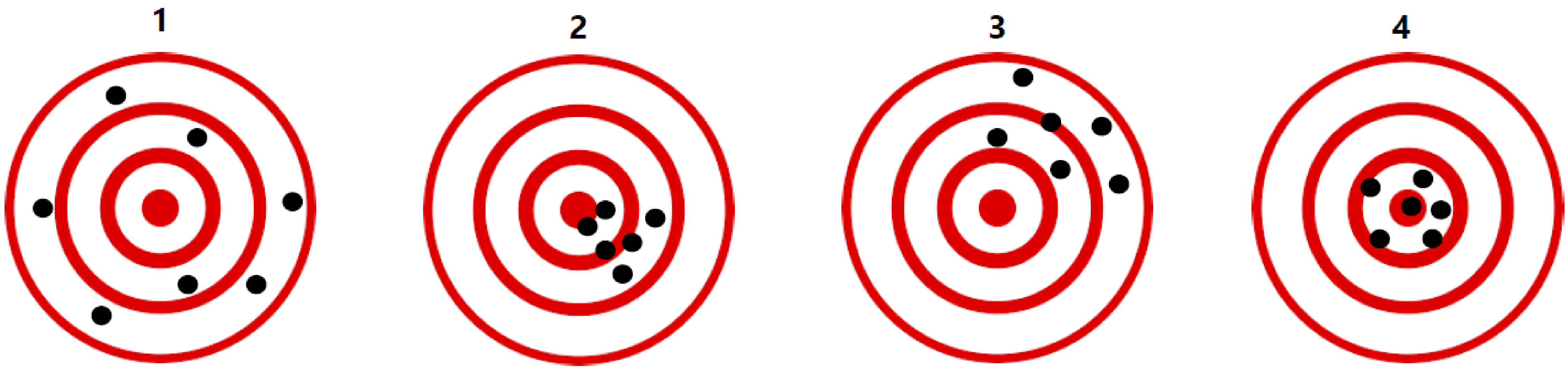
Forecast accuracy
- Backward looking, measuring unconstrained consensus FC accuracy
- Reflecting the accuracy of FC snapshot from x months in the past vs actual shipments of today
- # months to be adapted with the lead time of the product

Forecast bias
- Backward looking
- Reflecting unconstrained consensus FC deviation vs actuals, over-forecasting / under-forecasting
- Trend monitoring

Forecast consistency
- Forward looking
- Reflecting month on month FC fluctuation
- Trend monitoring

All KPIs can also be used for other forecast.

Accuracy and bias



Can you match the options with the pictures above?

Biased, low accuracy

Not biased, high accuracy

Biased, high accuracy

Not biased, low accuracy

Demand forecast KPIs

Forecast accuracy with MAPE:

i: *product customer combination*

ACT_i: *ACT qty of product customer combination* i

$$MAPE_i = \frac{|FC_i - ACT_i|}{ACT_i}$$

$$FCACC_i = Max\{0, (1 - MAPE_i)\}\%, \textit{ clipped to 100\%}$$

$$Timelag = x\ Months$$

Forecast bias

i: *product customer combination*

$$Bias_i = \frac{(FC_i - ACT_i)}{ACT_i}\%, \textit{clipped to} \pm 100\%$$

Forecast consistency

i: *product customer combination*

$$Bias_i = \frac{(FC_i(t) - FC_i(t-1))}{FC_i(t-1)}\%, \textit{clipped to} \pm 100\%$$

Targets need to be set for each

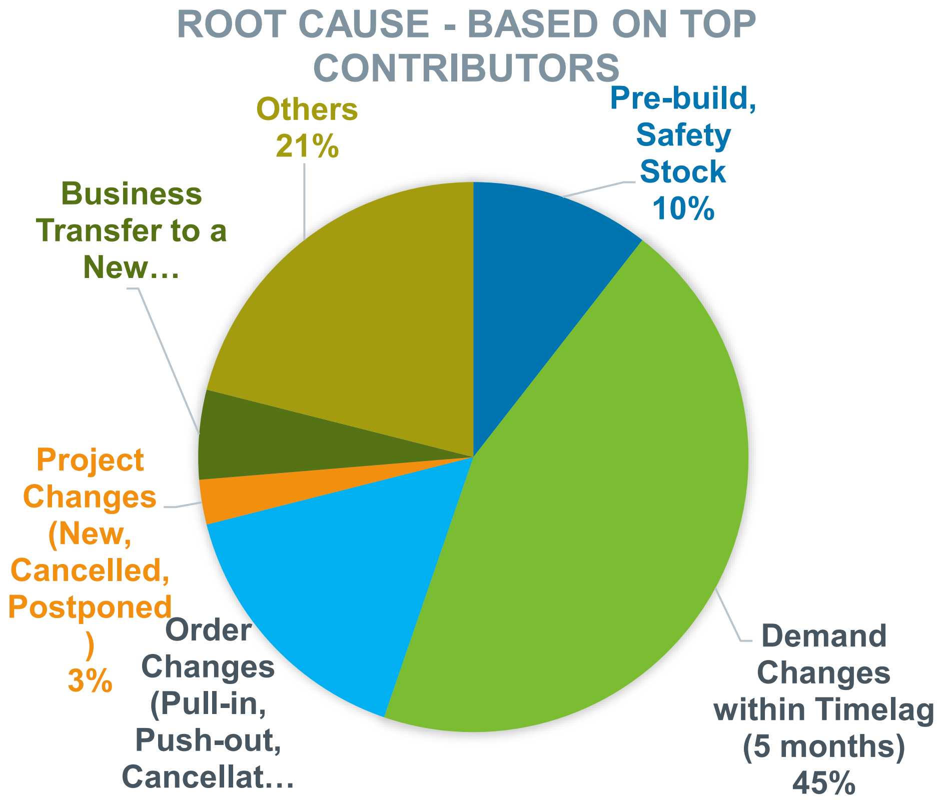
Forecast KPIs example

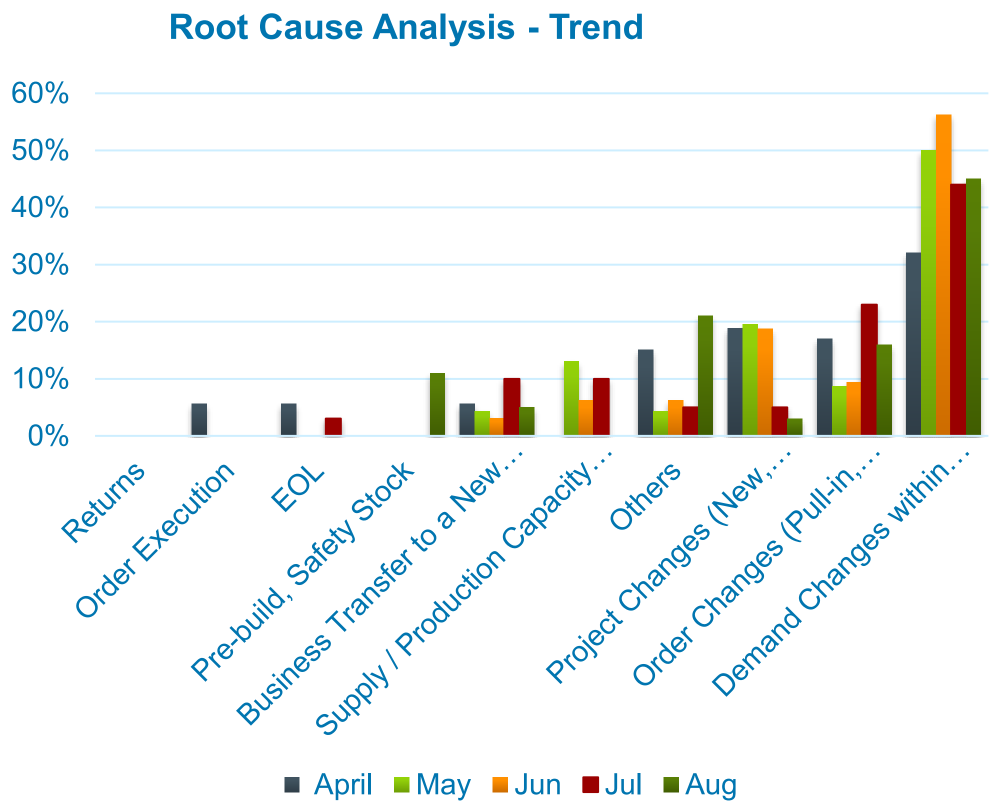
FC ACC	Q4'19			Q1'20			Q2'20			Q3'20		
Segment	Oct	Nov	Dec	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep
AUT	65%	67%	57%	50%	66%	72%	34%	43%	52%	52%	56%	50%
COM	32%	30%	29%	26%	37%	52%	62%	62%	46%	39%	49%	51%
COMP	0%	0%	100%	0%	0%	100%	100%	100%	0%	-	-	-
CON	68%	4%	4%	79%	15%	64%	33%	75%	39%	80%	35%	85%
FSF	36%	65%	34%	57%	50%	56%	40%	24%	60%	69%	49%	31%
IND	53%	50%	41%	54%	52%	39%	32%	58%	57%	55%	53%	46%
MED	65%	35%	56%	55%	40%	50%	48%	22%	46%	38%	59%	64%
No Mkt Seg	28%	56%	62%	26%	67%	23%	75%	12%	38%	92%	40%	45%

FC Bias	Q4'19			Q1'20			Q2'20			Q3'20		
Segment	Oct	Nov	Dec	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep
AUT	13%	30%	22%	-19%	17%	-5%	100%	100%	20%	63%	38%	-25%
COM	-39%	-43%	-10%	-19%	-12%	-16%	15%	24%	-19%	-3%	7%	-5%
COMP	-100%	-100%	0%	-100%	-100%	0%	0%	0%	-100%	-	-	-
CON	15%	-53%	-78%	75%	49%	33%	64%	15%	40%	2%	35%	53%
FSF	-41%	100%	100%	12%	7%	27%	59%	-37%	-16%	74%	-12%	38%
IND	7%	32%	-38%									
MED	45%	42%	52%									
No Mkt Seg	-53%	37%	-8%									

FC Consistency 0-5M	Q4'19			Q1'20			Q2'20			Q3'20			Q4'20
Division	Oct	Nov	Dec	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct
Corporate	2%	9%	4%	-3%	13%	8%	17%	17%	6%	-9%	1%	5%	9%
DIV 1	5%	1%	-8%	5%	-7%	-5%	8%	1%	-3%	-5%	1%	8%	-18%
BL1	38%	11%	24%	-56%	-8%	-4%	27%	2%	30%	-13%	-6%	-10%	-56%
BL2	4%	-3%	-10%	26%	-7%	-5%	6%	1%	-8%	-4%	2%	12%	-12%
DIV 2	-7%	-1%	7%	43%	24%	12%	22%	24%	10%	-11%	1%	4%	26%
BL1	-2%	-28%	-6%	0%	-7%	-4%	-5%	-16%	-5%	-4%	-6%	-6%	0%
BL2	-28%	-27%	5%	-31%	-1%	-5%	-22%	-16%	-19%	-24%	-24%	5%	32%
BL3	0%	23%	0%	13%	28%	14%	25%	29%	12%	-12%	0%	5%	27%
BL4	10%	18%	9%	3%	2%	7%	25%	2%	7%	10%	21%	-12%	14%

Forecast accuracy KPI into action





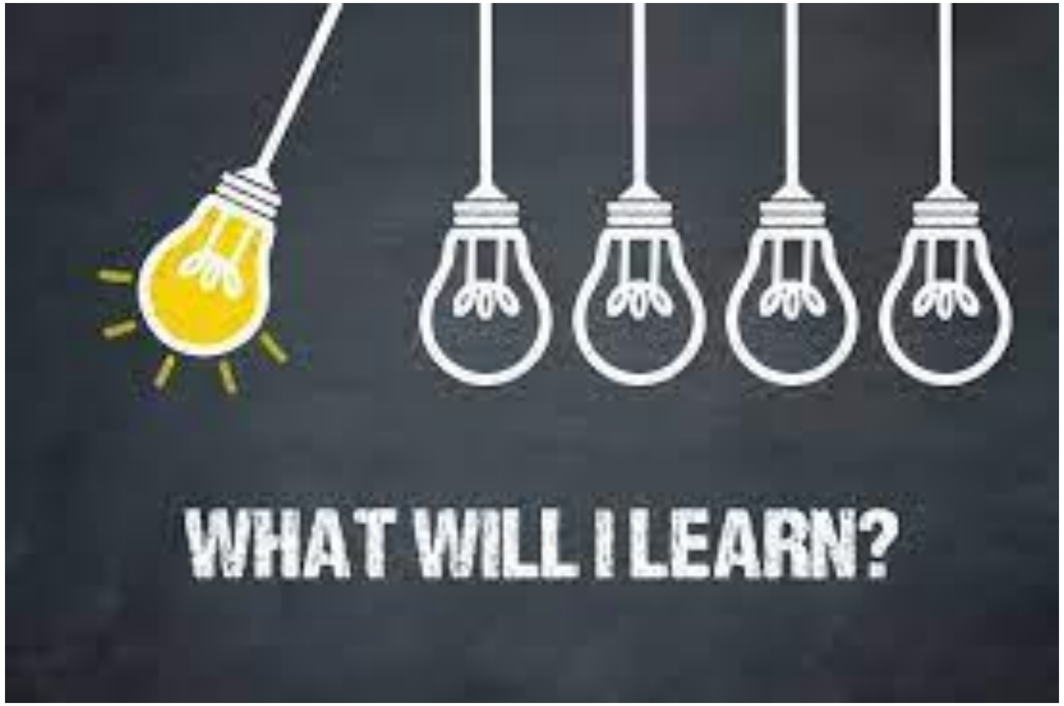
Wrap-up

- Though not 100% accurate forecasting is a very important and critical step for MSO
- There are several forecasting techniques to choose from, the most advanced is not mandatorily the best
- Actions driven out of forecasting KPIs help to improve the quality of the foreacst

Agenda Course 2

- Demand forecasting
  - Why forecasting demand
  - How to forecast demand
  - How to measure the quality of the forecast
  - o9 Guest speaker on ML and AI in forecasting
- Aggregate Planning
  - Planning Horizons
  - Aggregate Planning Problem
  - Unicorn case study

Learning Objectives



Understand the different planning horizons

Review the different leverage for developing a plan

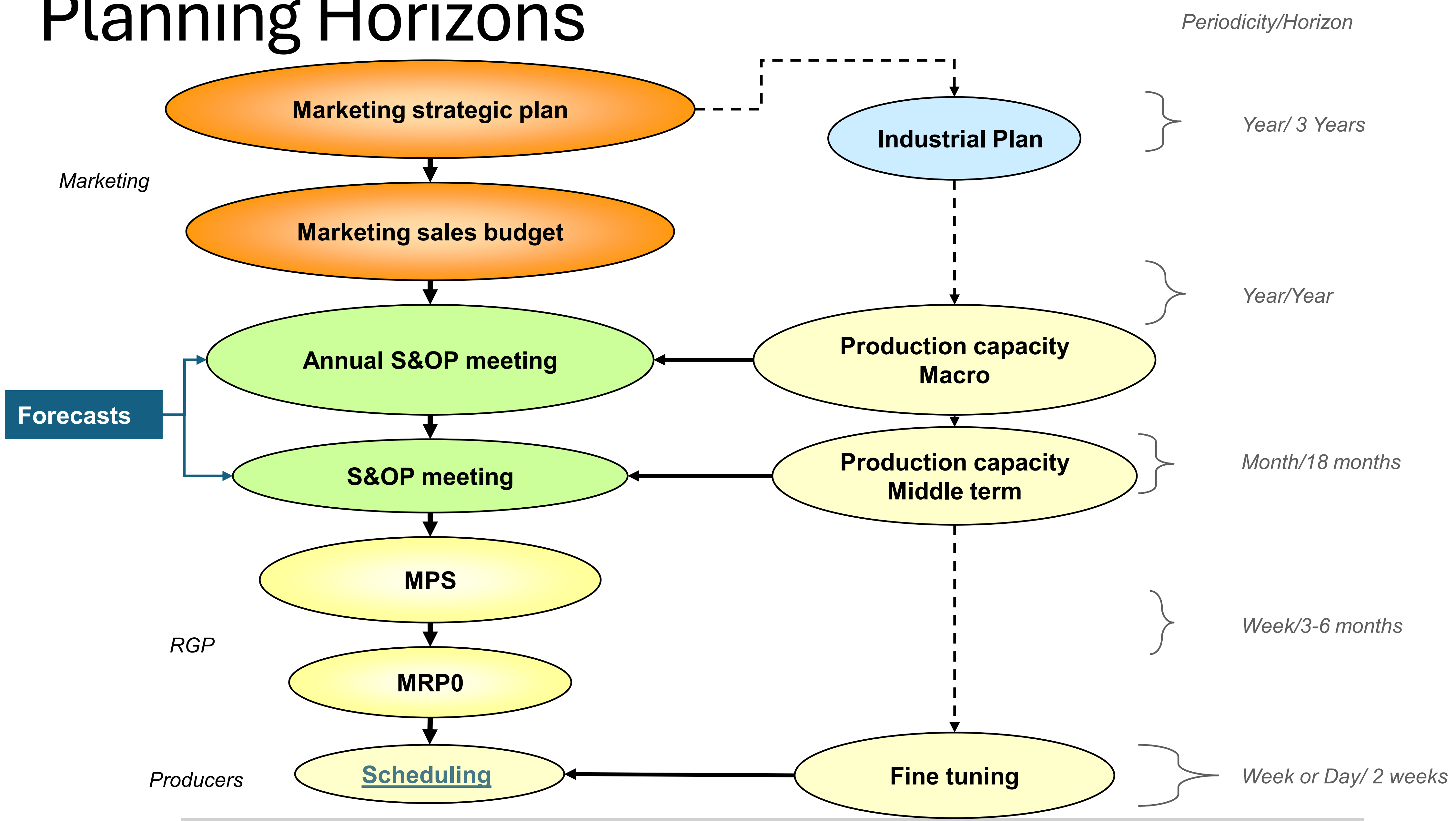
Solve an aggregate planning problem

Run and Cost various aggregate planning scenarios

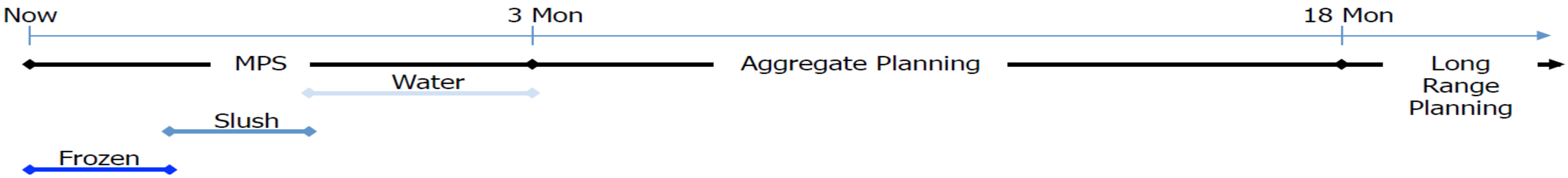
Link the outcome with revenue planning

« If you don't know where you are heading to, you end up elsewhere »
« Fail to plan plan to fail »

Planning Horizons



Planning Horizons



- The planning time fence (PTF) is an horizon where high level of details is needed, this is the time of your Master Production Schedule (MPS)
- The frozen fence is an horizon within the PTF where changes are not allowed.
- Outside the PTF planning should be done at an appropriate aggregated level.
- Benefits of aggregation : less volatility of the forecast
- Some timing:
  - PTF is about 3 months
  - Frozen fence is about 4 weeks
  - Within the PTF some more refined guidance on fluctuations can be given:
    - 2 weeks out of frozen fence +/-10%
    - 2 to 4 weeks after frozen fence +/-20%
    - etc

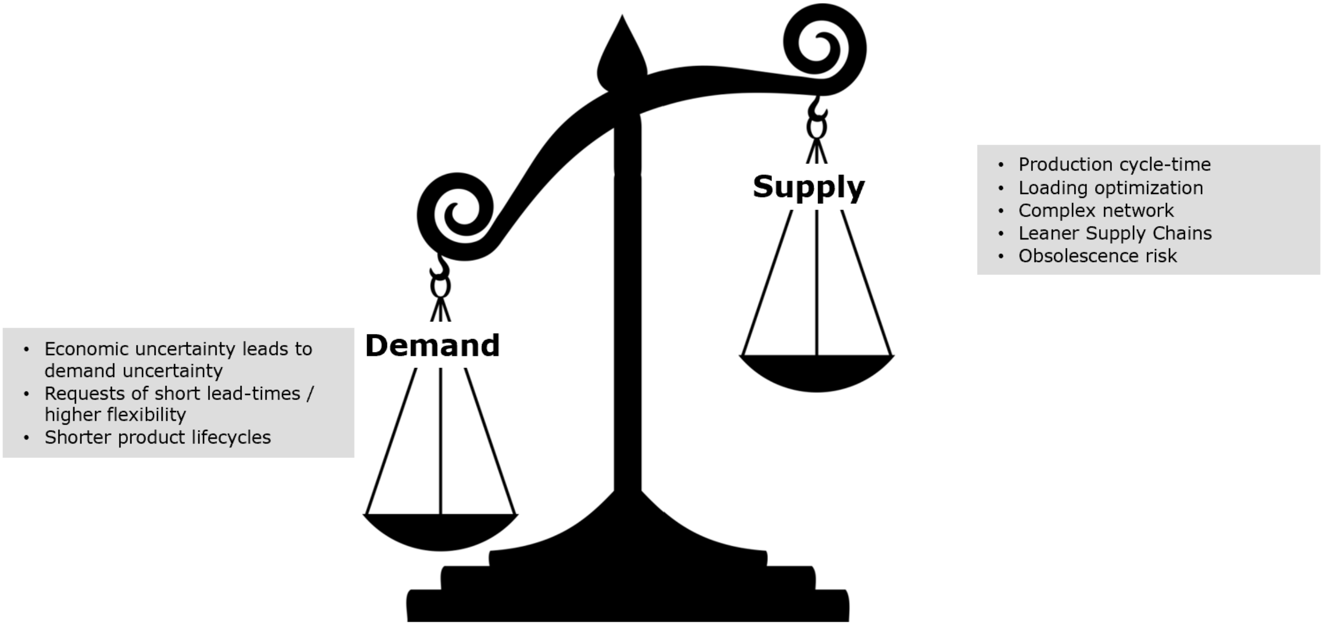
Aggregate Planning Premises

- Demand is known
- Horizon is fixed
- Aggregation strategy is known, the aggregate planning is NOT done at SKU level
- Cost is known
- Constraints are known (bottleneck)

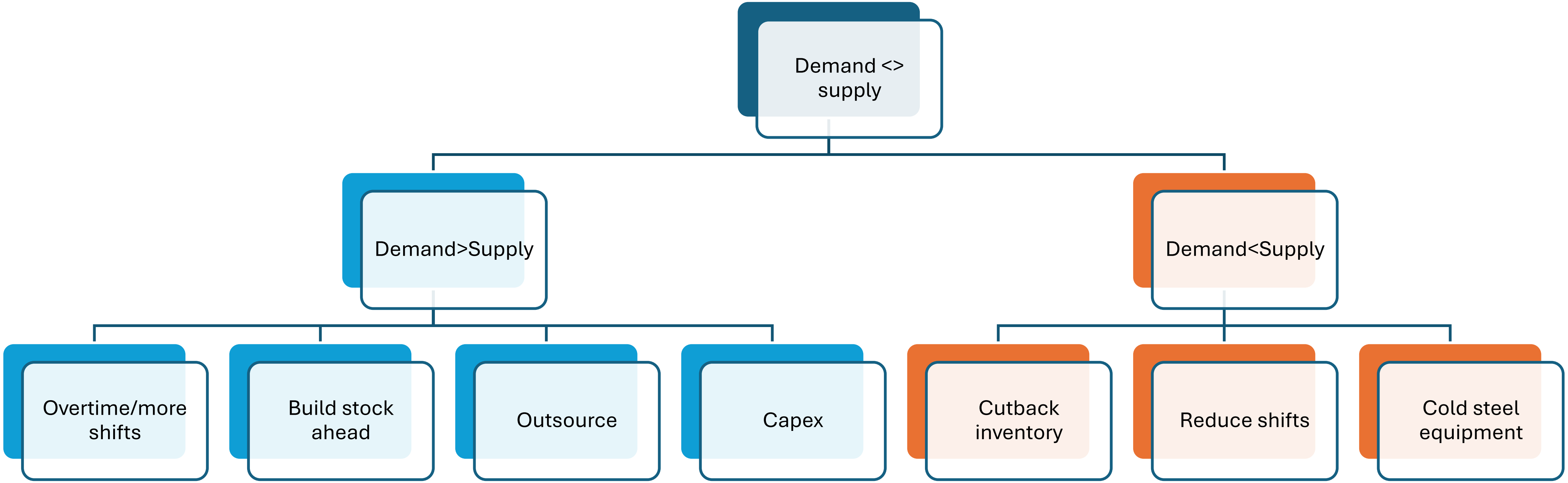
Aggregate Planning

Output:

- A feasible aggregate plan of :
  - What needs to be produced, where and how much
  - Where investment/divestment are needed
- An input for the revenue forecast
- A call for action when demand is lower than supply or the reverse



Type of decisions



Benefits of aggregation

When possible always consider to aggregate as it improves the forecast accuracy.
There are different options for aggregation. The most meaningful aggregation should be selected.
High level of aggregation :
Pros:
- Less noisy
- Easier to collect

Cons:
- Lose patterns at lower level
- Gap during the execution phase if the aggregation buckets contain items which behave very differently

Aggregation Strategy

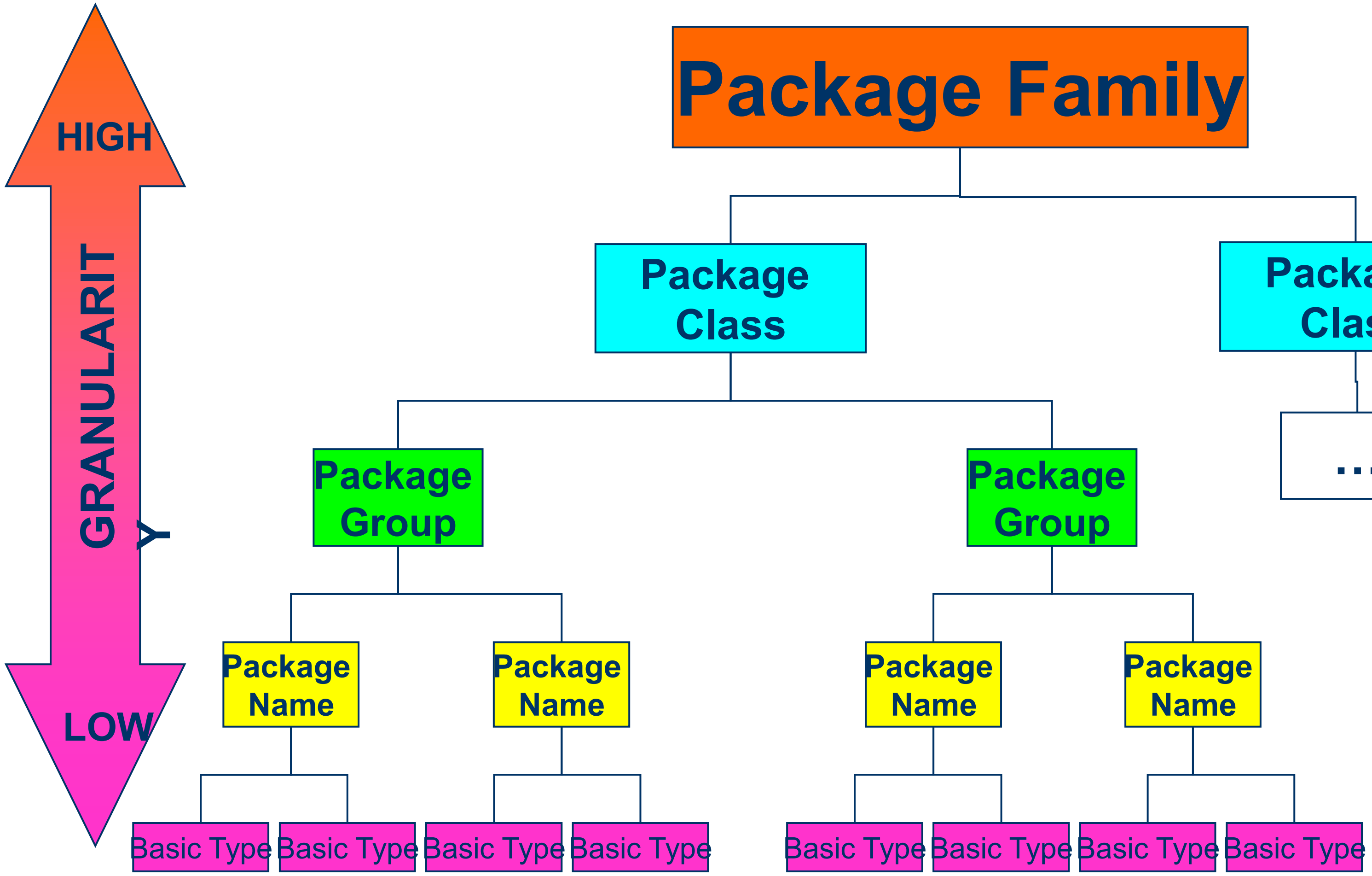
- Segment
- Style
- Size
- Colour

- Product Family
- Brand
- Package Size

- Model
- Colour
- Key options
  - Interior
  - Exterior
  - Wheel rim

- Product Family
  - Coffee
  - Tea
- Sub Family
  - Coffee Bean
  - Tea Flavour
- Key Ingredients

What are granularities?



Reference:**GRANULARITY DEPENDENCY OF FORECAST ACCURACY IN SEMICONDUCTOR INDUSTRY**
Holly Claudia OTT (Ph.D.)*, Stefan Heilmayer** and Cindy Sng Sin Yee***
ISSN 2083-4942

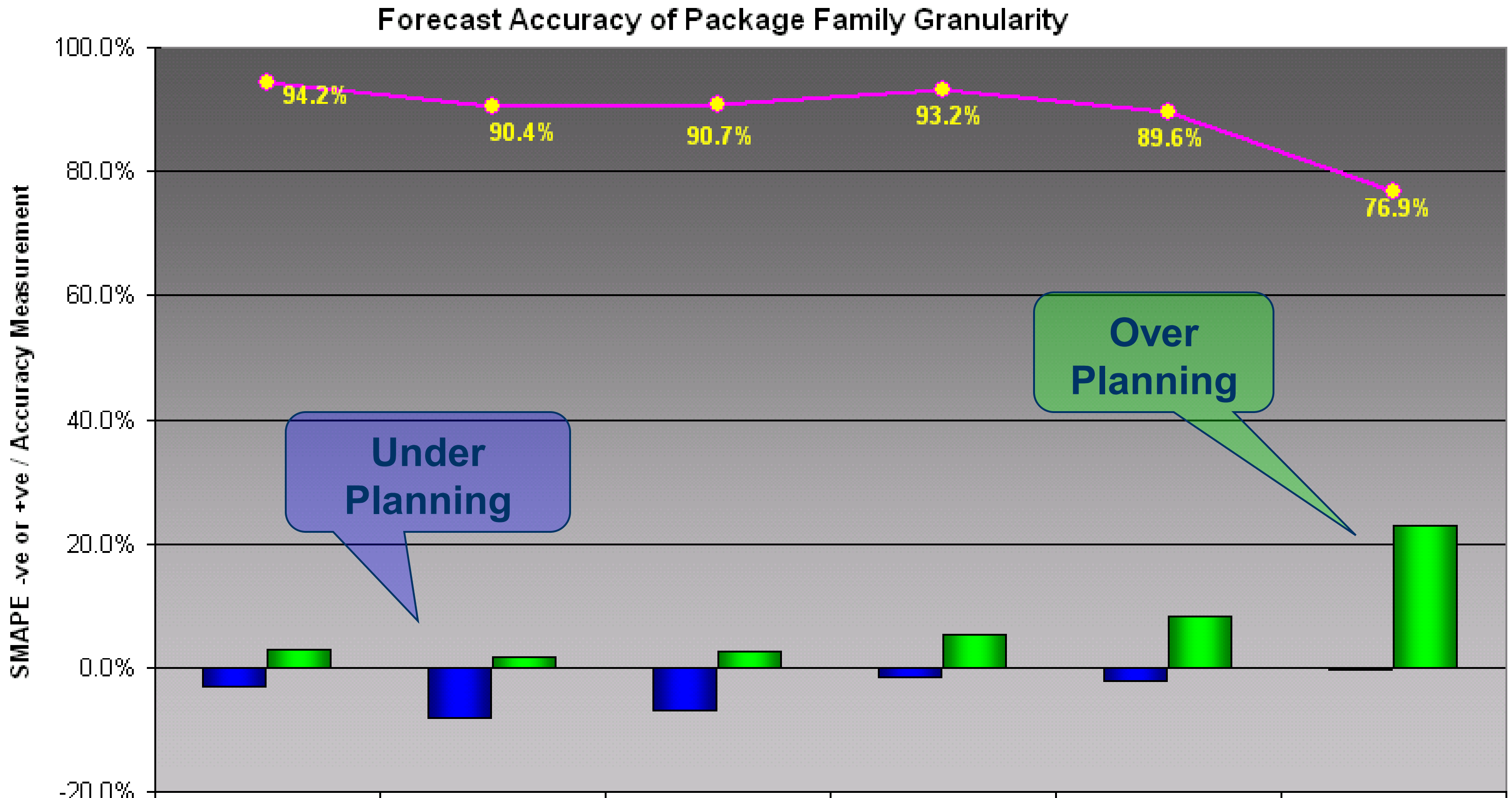
Summary of data & methodology

- ❑ **18 months data** (**Target month: 2011/JULY**)
- ❑ **Granularities of Power segment**
  - ➢ Basic types = 3394
  - ➢ Package Names = 422
  - ➢ Package groups = 174
  - ➢ Package class = 35
  - ➢ Package families = 16
- ❑ **Error formula used (SMAPE 3)**

$$\frac{\sum_{t=1}^{n} |O_t - F_t|}{\sum_{t=1}^{n} (O_t + F_t)} [\%]$$

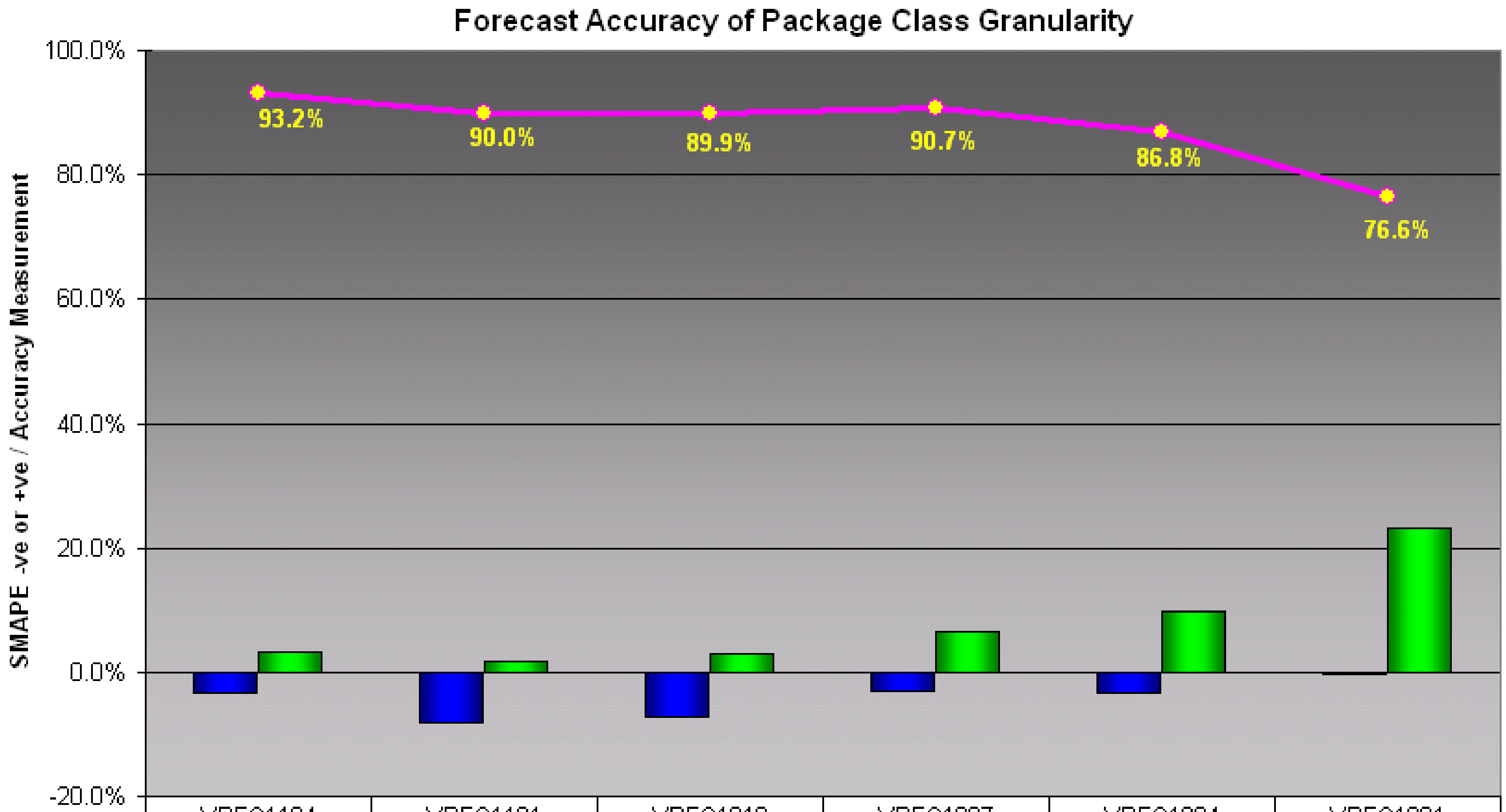
VRFC Accuracy = 100% - SMAPE3

Analysis results of 16 Package Families



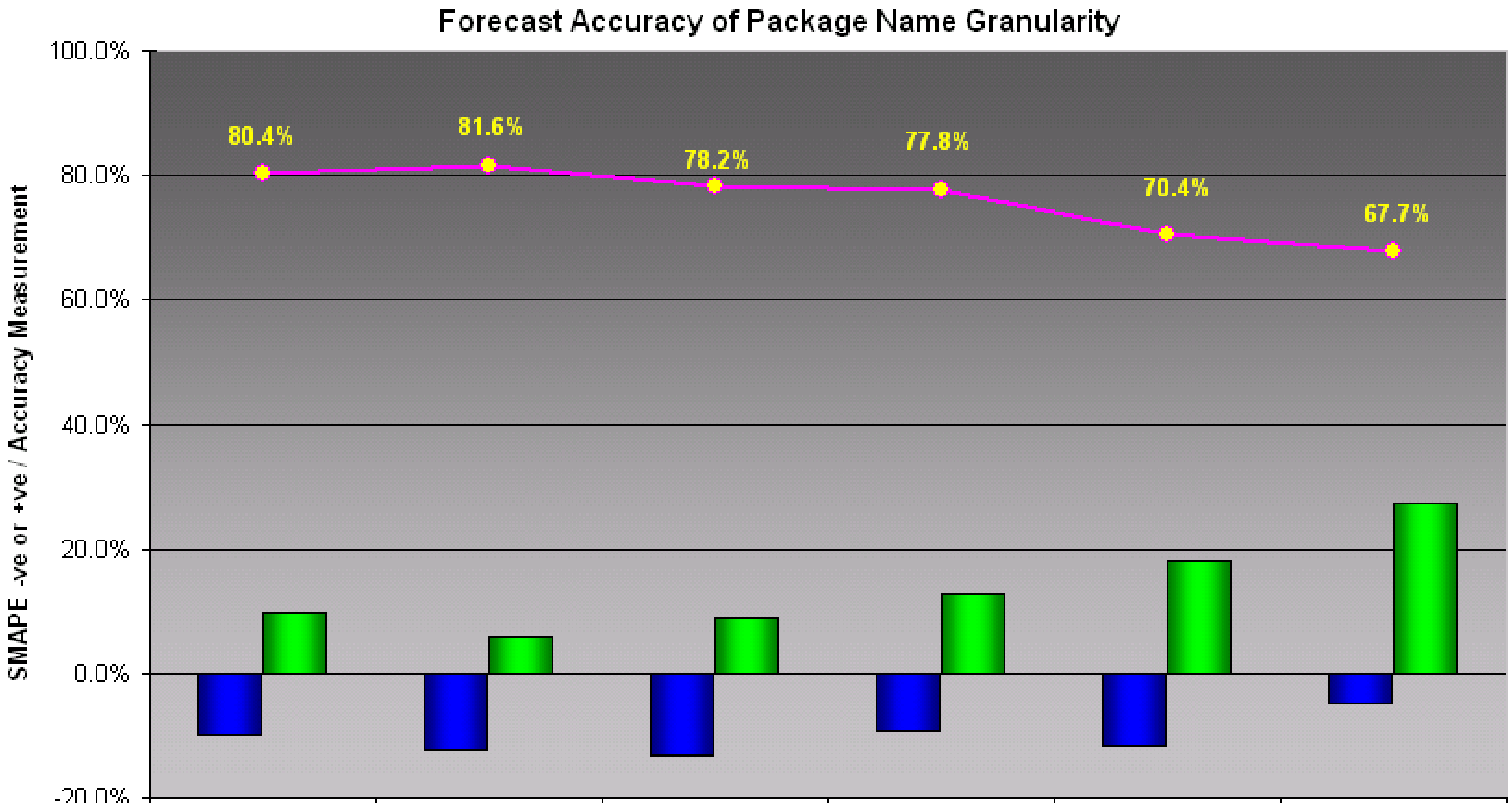
	VRFC1104	VRFC1101	VRFC1010	VRFC1007	VRFC1004	VRFC1001
SMAPE N	-2.9%	-8.0%	-6.8%	-1.6%	-2.0%	-0.2%
SMAPE P	2.8%	1.6%	2.5%	5.2%	8.4%	22.9%
SMAPE 3	5.8%	9.6%	9.3%	6.8%	10.4%	23.1%
Accuracy	94.2%	90.4%	90.7%	93.2%	89.6%	76.9%

Analysis Results of 35 Package Classes



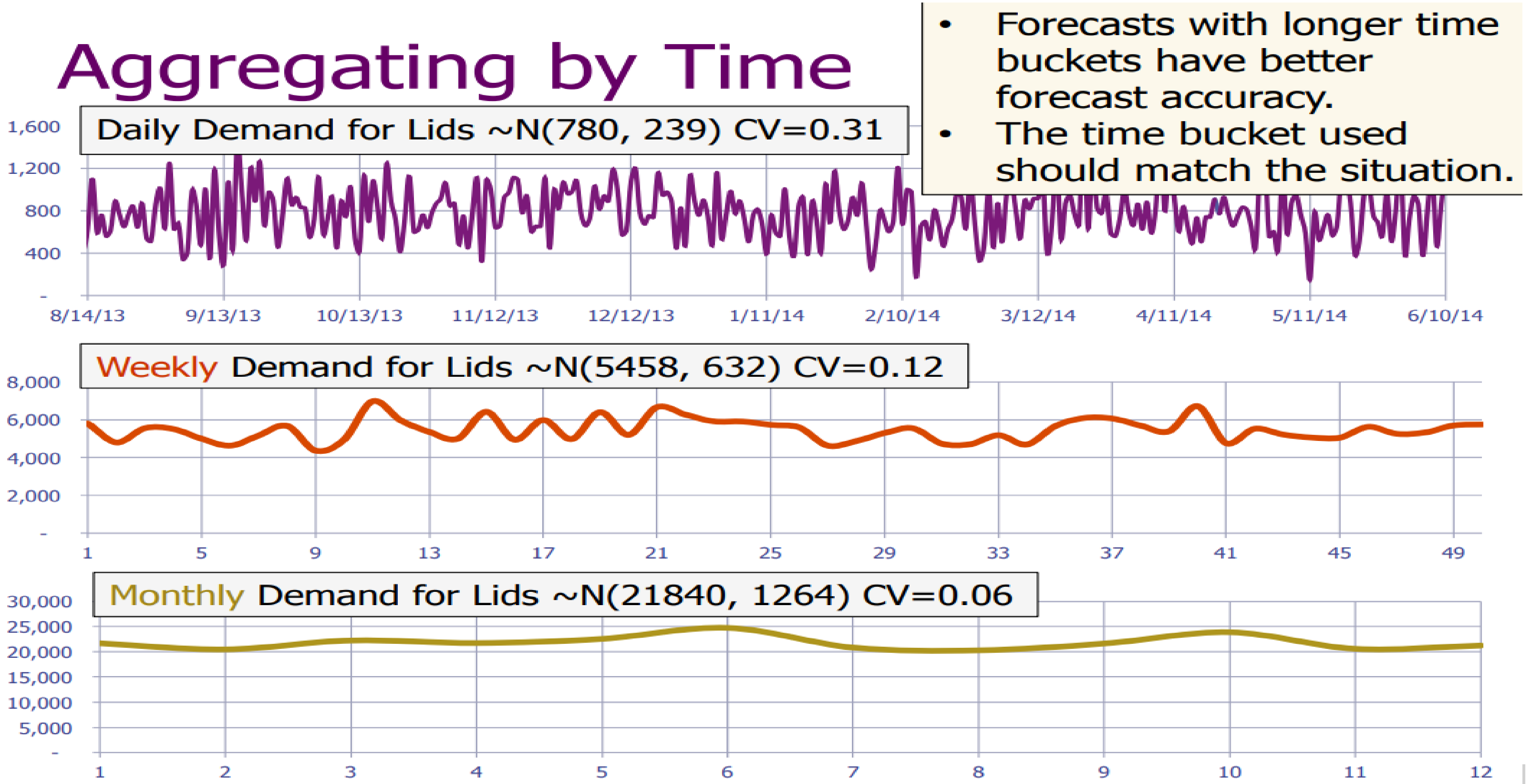
	VRFC1104	VRFC1101	VRFC1010	VRFC1007	VRFC1004	VRFC1001
SMAPE N	-3.4%	-8.2%	-7.2%	-2.9%	-3.4%	-0.4%
SMAPE P	3.3%	1.8%	2.9%	6.5%	9.8%	23.0%
SMAPE 3	6.8%	10.0%	10.1%	9.3%	13.2%	23.4%
Accuracy	93.2%	90.0%	89.9%	90.7%	86.8%	76.6%

Analysis Results of 422 Package Names



	VRFC1104	VRFC1101	VRFC1010	VRFC1007	VRFC1004	VRFC1001
SMAPE N	-9.8%	-12.3%	-13.0%	-9.3%	-11.6%	-4.8%
SMAPE P	9.7%	6.0%	8.8%	12.9%	18.0%	27.5%
SMAPE 3	19.6%	18.4%	21.8%	22.2%	29.6%	32.3%
Accuracy	80.4%	81.6%	78.2%	77.8%	70.4%	67.7%

Aggregate by Time



Aggregate Planning Problem

- Purpose:
  - determine production rate, workforce, inventory, outsourcing
  - Come-up with a feasible aggregated plan to meet demand
- Goal:
  - Maximize the firm profit over the horizon
- Leverage:
  - Workforce (overtime or temporary hire)
  - Production rate (chase vs level model)
  - Inventory
  - Backlog (overdue demand)
  - Outsourcing
  - New manufacturing site or new equipments
  - Product flexibility (postponement)

Unicorn Industry case study

Unicorn is a UK company with a manufacturing plant in Bristol. It manufactures pieces of equipment used in skiing gears. Its peak season is in winter while spring time tends to be rather quiet. They have many different SKUs but have aggregated them all in one flow for the purpose of our study. All costs have been standardized as well.

They want us to run various scenario and assess the best aggregate plan for the next 12 months which would minimize their cost.

The levers we have are :

- Size of the workforce (hiring and firing at a cost)
- Outsourcing
- Overtime at a cost
- Backlog
- Internal production level

We will develop different scenarios and assess which one is the most suitable for Unicorn.

Unicorn Industry case study

$	75	Material cost ($/unit)	160	Hours worked By employee per month
$	25	Inventory holding costs ($/unit/month)	10	Max overtime (hours/month/employee)
$	50	Cost of stockouts ($/unit/month)	500	Starting inventory
$	1,200	Cost for hiring and training new worker	500	Ending inventory (min)
$	3,600	Cost for laying off an employee	0	Backlog from previous period
	4	Labor hours required per item produced	0	Allowable ending backlog (max)
$	2,400	Regular worker cost ($/month)	36	Starting workforce
$	25	Overtime cost ($/hour)	30	Ending workforce (min)
$	175	Outsourcing cost ($/item)	36	Ending workforce (max)

The outsourcing cost includes material.

Overtime is at a max of 10 hours per employee. This limits the available capacity

Hiring up to 7 employees.

There are 4 weeks per month and 40 hours work time per week.

The demand is as per the attached table.

January 1	2,800
February 2	2,800
March 3	1,000
April 4	920
May 5	780
June 6	950
July 7	1,050
August 8	1,200
September 9	2,000
October 10	2,500
November 11	3,000
December 12	2,800

Unicorn Industry case study

Time Period	Inventory (#)	Workforce	Employees Hired	Employees Fired	Overtime (hrs)	Backlogged (units)	Internal Production (units)	Outsourced Production (units)		Monthly Demand Forecast (units)		Maximum Production (units)	Maximum Overtime (hours)
1	0	36	0	0	0	2 300	0	0	≥	2 800		1 440	360
2	0	36	0	0	0	5 100	0	0	≥	2 800		1 440	360

Yellow cell: for you to key your entries

Unicorn Industry case study

With the information you have :

1) Understand the provided spreadsheet to assess the total cost based on various production scenario
2) Can Unicorn fulfil the demand without relying on outsourcing and without hiring/firing anybody? Does this plan meet the criteria?
3) How does the plan look like if they hire 7 employees and are still not leveraging on outsourcing ? Does this plan meet all the criteria?
4) What happens to the cost if they cannot fire/hire/use overtime but can use outsourcing? Is this the way to go ?
5) Any other smart option we should investigate ?
6) Draw your summary and conclusions

Aggregate Planning: more complex models

- So far our constraint was linked to human resources
- In reality constraints might come from equipment, space, material, ...
- For material we will look into this when discussing MRP
- For equipment a bill of resources need to be established ie how much time is needed on the equipment to produce a product family

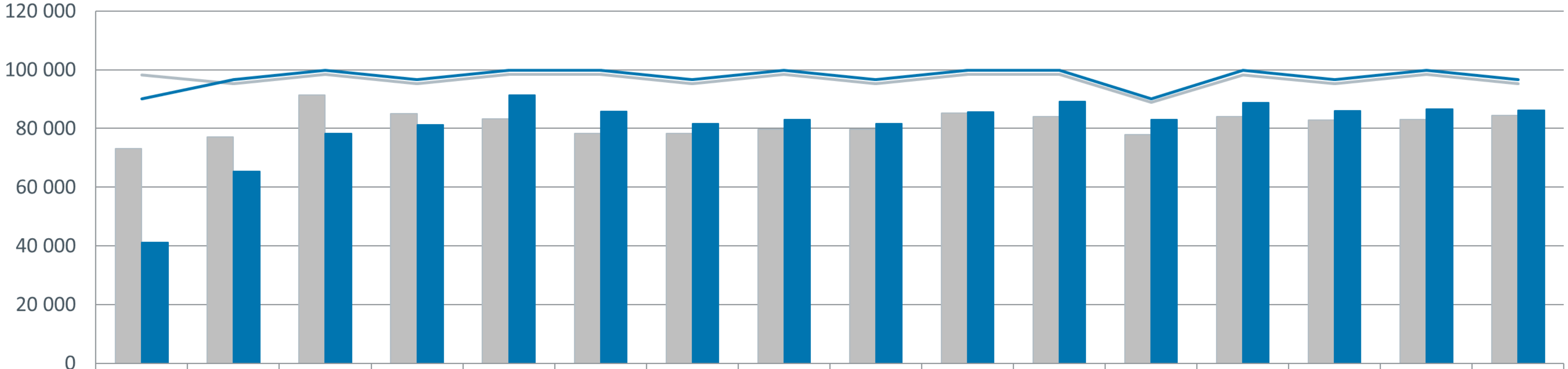
Resources	Product A	Product B	Product C	
Dicing (min)	2.1	1.9	1.3	Resource Planning
Die Bonding (min)	1	1.5	0.5	Resource Planning
Mold machine (min)	2	1.2	1.1	Resource Planning
Test (hrs)	3	1.9	2	Resource Planning

Bill of Resources analysis

- In our example we have a snapshot for one month, in reality you would look at this over the entire planning horizon.
- If products in a family are not similar enough your analysis might show no issue in the aggregate planning run but issues in the MPS run

Key Resources	**Demand Plan per product**			**Capacity**		
	A	B	C	Requested	Installed	Stretched
Description	10000	20000	15000	45000	50000	50500
Dicing (min)	21000	38000	19500	78500	79000	80000
Die Bonding (min)	10000	30000	7500	47500	47500	48000
Mold machine (min)	20000	24000	16500	60500	100000	110000
Test (min)	30000	38000	30000	98000	100000	110000

Bill of Resources example



	03.2021	04.2021	05.2021	06.2021	07.2021	08.2021	09.2021	10.2021	11.2021	12.2021	01.2022	02.2022	03.2022	04.2022	05.2022	06.2022
Consumption M-1 H	73 219	77 040	91 482	85 075	83 209	78 219	78 232	79 732	79 640	85 198	84 065	77 847	84 004	82 794	83 139	84 380
Consumption M H	41 156	65 332	78 230	81 212	91 454	85 886	81 679	83 032	81 674	85 692	89 298	83 136	88 804	85 984	86 722	86 333
Capacity M-1 H	98 308	95 170	98 342	95 170	98 342	98 342	95 170	98 376	95 170	98 342	98 342	88 825	98 308	95 170	98 342	95 170
Capacity M H	90 161	96 638	99 860	96 638	99 860	99 860	96 638	99 894	96 638	99 860	99 860	90 196	99 825	96 638	99 860	96 638

Bill of Resources example

	Utilization M	Utilization M	Utilization M	Utilization M	Utilization M	Utilization M	Comment
Quarter	1.2021	2.2021	3.2021	4.2021	1.2022	2.2022	
Resource							
PS Tester - EX	52	66	72	79	75	73	
PS Tester - SM	40	58	81	90	146	224	
[-] PH Tester LX DX EX M	47	88	102	94	103	102	
PH Tester - SM	24	68	111	98	100	155	
PH Tester - DMX	54	100	94	89	79	78	
PH Handler - Prober	7	90	95	102	109	111	
PH Handler - MT95XX	44	70	78	85	121	118	
PH SAW	15	96	97	99	93	92	

How many machines?

- The company Good Foot Sandals asked you to identify the required capacity for one of their critical operation, given the expected demand. The capacity is measured in number of machine hours per year. Each machine is operated by a team.
- The company manufactures 3 products: male sandals, female sandals and kid sandals.
- Below table shows the process time (manufacturing and prep), the lot size and the forecasted demand
- The company has currently 1 team per day, 8h per day and 5 days per week, 50 weeks per year
- Their experience tells them that they should reserve 5% of capacity for maintenance

Product	Manufacturing (h/pair)	Prep (h/lot)	Lot Size (pairs / lot)	Forecasted pairs per year
Sandals male	0,05	0,5	240	80 000
Sandals female	0,10	2,2	180	60 000
Sandals kid	0,02	3,8	360	120 000

Wrap-up

The aggregate planning (also called S&OP):

- Is done at an aggregate level
- Covers 18 months horizon
- Balances demand and supply
- Triggers investment decision or revenue catch-up decisions
- Is an input for the revenue forecast communicated more broadly
- Aligns all parties towards the same goal
- Is reviewed every month

THANK YOU

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TECHNOLOGY AND DESIGN